AERead: Evaluating Economic Decisions in Interactive Environments

Chenyu Li · Zeyu Sun

contact@aeread.org

Research working paper · 18 October 2026 · Preliminary results; feedback and contributions welcome

Abstract

AI agents are increasingly asked to make economic decisions for a principal, such as whom to trade with, what to verify and when to commit. Here a completed transaction can destroy value and a refusal can be correct, so task-completion or agreement scores cannot tell whether an agent served its principal. We present AERead, an environment that scores these decisions by the value an agent obtains. Each case defines a distribution of economic worlds, seats with private information, counterparts with their own interests, and a reference policy whose access to information is stated, so that a shortfall separates poor decisions from information the agent never had. One runtime executes every case and recomputes each score from the sealed record without a model; quality control runs over populations of worlds, not single tasks. We build cases in housing, procurement, data-center contracting and refunds on this runtime. The counterpart is part of the measurement: one agent's earnings lead over another is 42% of the value available in a deal head to head and 7% against the same scripted counterpart. Shortfalls are broken down by stratum and by action class; in housing, nearly all of the gap to the reference falls in worlds where the tenant's favourite listing is a lemon. Of 25 defects selected from our quality-control record, 14 were invisible in any one world. Housing now passes four of the five quality-control gates and no case passes all five, so the results compare named agents on stated worlds rather than rank models. AERead's code is open source: a new case supplies its economic rules, worlds and reference, and inherits the runtime, replay and the quality-control standard.

1. Introduction

An agent buying on behalf of a firm must decide more than whether it can place an order. It must choose whom to trust, whether another test is worth its cost, and when to commit before an offer expires. A tenant faces the same problem when inspecting a home; a negotiator faces it when exchanging price for contractual risk. In these settings, completing a transaction can destroy value, while refusing it can be the right decision. We use *economic-agent readiness* to mean an agent's ability to obtain value for a principal within a stated mandate, given limited information and other parties' interests.

Evaluating this ability requires a reference for what the agent could have achieved. A good realized outcome can follow a poor decision, and a shortfall from a fully informed optimum can reflect information the agent never had. The counterpart also matters: a strategy that succeeds against a cooperative seller may fail against a different policy. Existing work provides interactive task evaluation and economic decision benchmarks (Merrill et al. 2026; Fish et al. 2025; Zhang et al. 2026). The problem we address is how to make information, counterpart behavior and reference values explicit across different economic cases, then test whether the resulting scores measure the intended decisions.

AERead provides a framework for constructing and evaluating such cases. Each case specifies the parties, their observations and legal actions, the world distribution, and checks on behavior and outcomes. A shared runtime executes play and records evidence. Case-specific references give scores an economic interpretation: a feasible policy, an optimal policy under the deciding seat's information, or a labelled full-information upper bound. Results retain their native units and reference types.

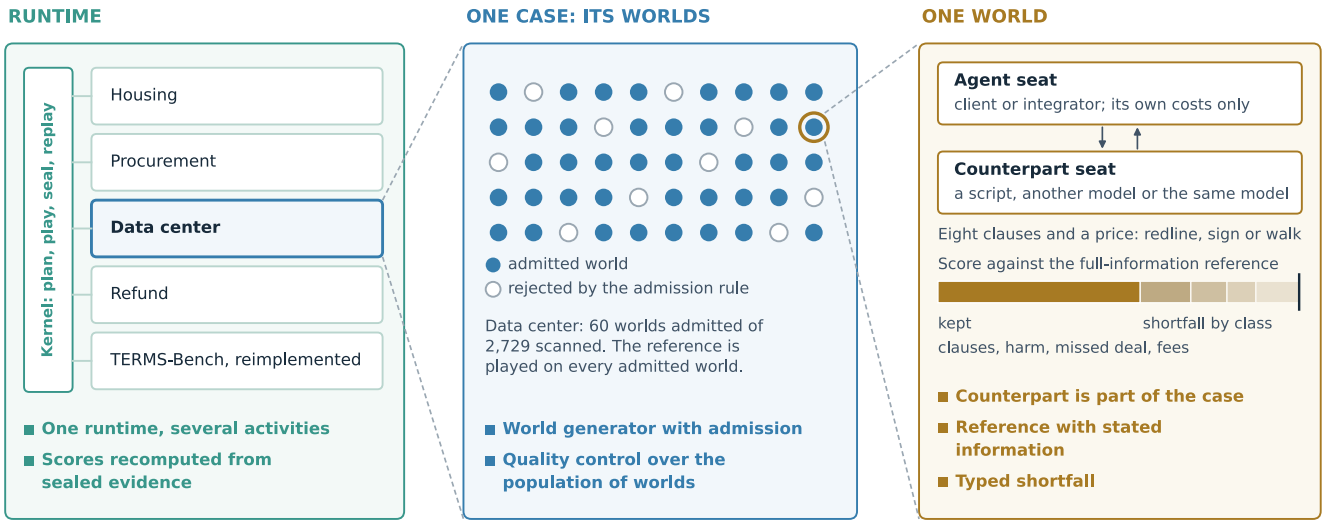


Figure 1.1. The three levels of an AERead evaluation, with the Data-center agreement as the example. The runtime hosts several case families (left). A case draws a population of worlds and admits those where the decision matters; quality control runs over that population (centre). In one world, the agent holds a seat, a counterpart holds another, and the score is compared with the case reference and split into shortfall classes (right). Dashed lines mean "contains", not a sequence. Each panel lists the properties of Section 1 that belong to its level. The dot pattern and the score bar are schematic.

AERead combines seven properties. Prior benchmarks have some of them, TERMS-Bench five in full; AERead has six in full and the seventh, a shortcut-screening admission rule, in three of its cases (Table 2.1). The first three concern a single world, the next two a case's population of worlds, and the last two the runtime (Figure 1.1). Each is built and was applied to the cases reported here.

1. **Value against a reference with stated information.** Scores are economic values in each case's own units, measured against a feasible policy, the best policy given the deciding seat's information, or a labelled full-information bound. Each reference is played through the environment on every admitted world (Section 6).

2. **The counterpart is part of the case.** Other parties are seats with their own information and interests, played by scripted policies, another model or the same model, and each claim names its counterpart. In the Data-center case, one agent's earnings lead over another is 42% of the value available in a deal head to head and 7% against a fixed scripted counterpart (Section 5).

3. **Typed shortfall.** The gap to the reference is split into observed action classes, such as tests not taken, worse terms, harmful agreements and negotiation fees, so a result states where value was lost as well as how much (Section 5).

4. **World generators with admission.** A case is a generator of worlds, and its packs regenerate from their seeds. An admission rule, computed before any agent plays, keeps only worlds in which simple shortcuts lose a stated margin: the Data-center comparison used 60 worlds admitted from 2,729 scanned. Housing's v15 pack admits a world only if the best informed tenant beats a simple inspect-then-sign rule by $25, and the reported Procurement pack only if a rule that plans one newcomer's test exactly pays the premium while the best policy beats paying it by 10%: 20 worlds from 5,164 scanned (Section 5).

5. **Quality control over populations of worlds.** Cases are qualified through typed gates; simple-rule ceilings are scored beside the agents and briefs are audited. Of 25 defects selected from our record, 14 were invisible in any single world (Section 6).

6. **Scores recomputed from sealed evidence.** The kernel hashes a plan before execution, seals the events, and replays each receipt to recompute its scores without calling a model; no model judges a score (Section 4).

7. **One runtime for several activities.** Families supply only the economic rules. Four maintainer-built cases and a reimplementation of three TERMS-Bench counterpart families run on the same kernel (Section 5.3).

A researcher studying another economic decision therefore writes only its rules, worlds and reference: execution, sealing, replay, counterpart seats and the quality-control gates come with the runtime, so a new case starts with the same evidence and checks as the existing ones. Sections 3 and 4 describe the case design and the runtime, Section 5 the cases, and Section 6 the quality control. The cases are maintainer-built worked examples; Housing passes four of the five qualification gates and none passes all five; Section 7 states what their results do and do not establish.

2. Related work

Table 2.1 compares the closest benchmarks on the seven properties of Section 1; the paragraphs below discuss them by theme.

Table 2.1. The seven properties in related benchmarks, read from each paper's description of its own method: ● present, ◐ partly present, – absent. Each cell was read against a written definition, and the lower mark was chosen where a reading was borderline; page references for every cell are kept with the source. AERead's row describes the cases reported here; Housing v15 passes four of the five qualification gates and none passes all five (Section 6). TERMS-Bench has five of the seven properties in full. AERead adds several activities on one runtime and, in three of its cases, an admission rule that screens shortcuts.

Benchmark	1. Value against a reference with stated information	2. Counterpart is part of the case	3. Typed shortfall	4. World generator with admission	5. Quality control over populations of worlds	6. Scores recomputed from sealed evidence	7. Several economic activities on one runtime
Terminal-Bench	–	–	◐	◐	●	◐	◐
τ²-bench	–	◐	◐	◐	◐	◐	◐
GDPval	–	–	◐	–	◐	–	◐
Agents' Last Exam	–	–	◐	◐	◐	◐	◐
EconEvals	●	–	◐	◐	◐	◐	●
NegotiationArena	◐	●	◐	–	–	◐	◐
AgenticPay	◐	●	◐	–	–	◐	◐
TERMS-Bench	●	●	●	◐	●	●	◐
AERead (this paper)	●	●	●	◐	●	●	●

Economically valuable work.

GDPval selects professional tasks by occupation and evaluates deliverables through expert comparison (Patwardhan et al. 2025). Agents' Last Exam (ALE) evaluates expert-authored professional workflows with task-specific verification (Sun et al. 2026). AERead evaluates decisions under a principal's mandate, with responsive counterparts and payoff references that specify the deciding party's information.

Interactive agent evaluation.

Terminal-Bench verifies final environment state and supplies human-written solutions (Merrill et al. 2026). AppWorld combines an application engine with state-based evaluation, including collateral-change checks (Trivedi et al. 2024). τ^2-bench lets agent and user act on shared state (Barres et al. 2025).

Economic decisions and negotiation.

EconEvals covers procurement, scheduling, pricing and economic choice (Fish et al. 2025). NegotiationArena studies bargaining tactics (Bianchi et al. 2024); SidConArena combines negotiation, production and auctions (Feng, Chen, and He 2026). AgenticPay evaluates buyer–seller bargaining under private reservation values, scoring agreement, surplus allocation and speed (Liu, Gu, and Song 2026). TERMS-Bench specifies counterpart types, policies and payoffs to diagnose negotiation through oracle-reference gaps (Zhang et al. 2026); our case specifications follow its decomposition into environment, protocol and counterpart policy (Appendix A), and Section 5.3 reimplements three of its counterpart families on AERead's runtime. AERead retains case-specific payoff units and distinguishes full-information bounds from information-constrained policy references, with QC of world distributions and measurement paths.

Shared environments and evaluation quality.

OpenSpiel and PettingZoo provide common interfaces for games and multi-agent environments (Lanctot et al. 2019; Terry et al. 2021). AERead similarly separates shared execution from case semantics (Appendix H). Terminal-Bench combines executable reference solutions, automated checks, human review and adversarial exploit testing (Merrill et al. 2026). The Agentic Benchmark Checklist identifies task and reward defects that distort performance (Zhu et al. 2025). Our QC analysis tests whether prompts, controls and counterpart assignments support valid comparisons across distributions of economic worlds.

---

3. Task design and construction

Three of AERead's properties are fixed when a case is designed: the reference and the information it may use, the counterpart seats, and the world generator with its admission rule.

3.1 Cases, worlds and seats

A *case* defines an economic decision problem; a *world* is one instance of that design, with particular valuations, private types, costs and deadlines. A *seat* is a party's position in the interaction, with its own objective, information and permitted actions. A policy occupies a seat and chooses actions from its observations and available history. The family implementation supplies the environment, observation rules, legal actions, transitions and verifier.

For example, Housing places a tenant among four listings with hidden quality. The tenant can inspect, make offers and decide whether to sign; landlord replies and outside demand affect the opportunities that remain. Inspection changes both information and cost. The verifier therefore needs to distinguish a legal action, a favorable realized outcome and a good decision given the tenant's information.

In a controlled comparison, the evaluated policy occupies one seat and counterparts follow fixed policies. Cross-play and self-play change those assignments and answer different questions. Each comparison names the seat, counterpart configuration and world distribution to which it applies. Appendices A–D give the case specifications, counterpart policies, references and measures.

3.2 Selecting economic decisions

We organize coverage around activities in which a party's choices affect the value of a transaction (Li 2026). A candidate case needs a consequential decision, a counterpart with its own interests and information, and an outcome that can be evaluated inside the world. The intended coverage spans business purchases, labour contracts, capital and corporate transactions, household purchases, and public purchases. Contractual terms, risk allocation and disputes enter as decisions within these transactions.

The four examples cover business and household purchases. They were selected to exercise different references and counterpart configurations; they are not a representative sample of economic activity. Economic stakes motivate which decisions to study, but the reported scores are case-specific quantities, without a cross-family aggregate or an estimate of economy-wide gains.

3.3 Constructing and qualifying a case

A contribution begins with a decision and an interpretation of success. The author specifies the principal's objective and constraints, what each seat can observe, which actions commit resources, and how counterpart policies respond. The brief must expose the rules needed to make the decision while preserving the intended private information. These choices define the mechanism that a score is meant to test.

The author then provides a world generator or fixed pack, an admission rule, and a reference whose information access is explicit. Admission screens each candidate world before any agent plays and before hidden types are drawn. Computing exactly from the public world and the declared prior, it keeps a world only if simple shortcuts, such as accepting the default terms or ignoring private information, fall short of the reference by a stated margin. Whether the reference can be reached is a separate check: executable controls play it through the actual environment and verifier on every admitted world, alongside simple policies that reveal shortcuts or ceiling effects that admission missed. The pack must also be valid, free of duplicates, and split into disjoint development and holdout ranges. A full-information bound is labelled separately from a policy the evaluated seat could execute. The case declares score units, direction and treatment of invalid actions and incomplete episodes.

Before a confirmatory comparison, the campaign freezes the worlds, seat assignments, prompts, agent settings, budgets and analysis. Its evidence package should connect each reported measure to the retained actions and outcomes, the scoring implementation, and a command that reproduces it. Section 4 describes the shared execution path; Appendix H records where the current drivers and headline analyses extend beyond the receipt's coverage.

This is the contribution and qualification standard. Implementation support and completed checks vary across the four examples, as Section 6 and Appendix F document. Cases can be useful for development before qualification, provided their reports identify missing checks, preserve failed attempts, and distinguish analyses declared before execution from later corrections. The current examples demonstrate the workflow and its failure modes; they do not establish successful adoption by external contributors.

---

4. Shared runtime and evaluation harness

The runtime provides two of AERead's properties: one kernel executes every family, and every score is recomputed by code from sealed evidence. The shared runtime, or *kernel*, executes the case designs in Section 3. Families supply the economic rules; the kernel schedules play and records the evidence used for evaluation (Figure 4.1).

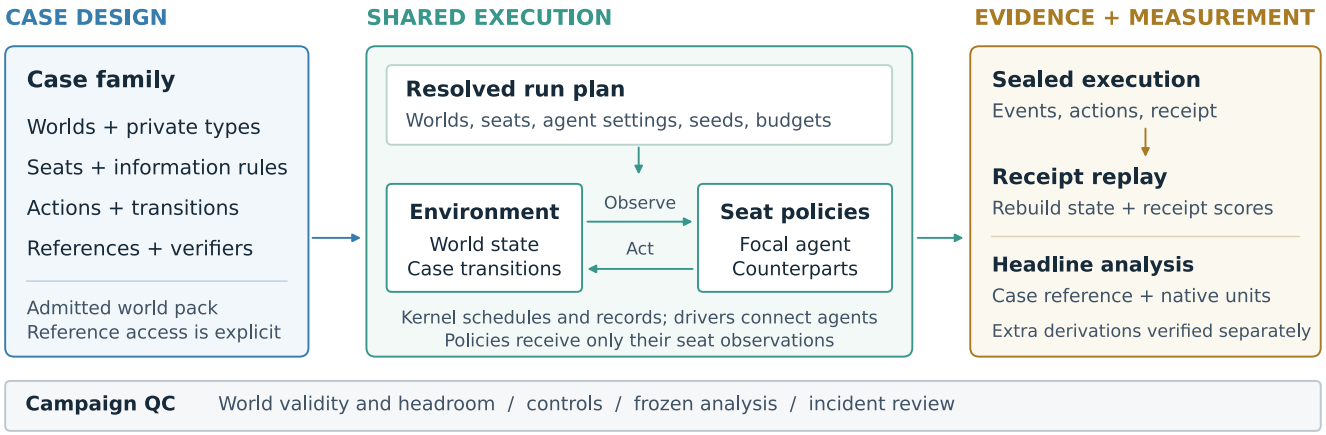


Figure 4.1. AERead architecture. Families define the economic world and reference; the runtime coordinates seat policies and retains execution evidence. Agents receive only their seat observations, while scoring uses explicitly declared reference access. Replay checks receipt consistency; separate headline derivations require their own verification. Applied coverage and QC verdicts vary by campaign (Appendices F and H).

4.1 Verification

Checks compare behavior or outcomes against five kinds of reference: an *answer key* specifies a correct resolution; a *rule* specifies a constraint; *value* measures economic outcomes against a policy or bound; *relative* compares named agents or populations; and a *rubric* rates an artifact. Each check states its output (pass/fail, count or amount) and whether it concerns the outcome or the path. Results retain their native units, without a cross-family aggregate. Value references distinguish a reachable policy, the best policy given the decider's information, and a full-information upper bound; the last need not be attainable by the evaluated seat.

The case-study scores are computed by code. Kernel replay reconstructs state and each receipt's score from recorded actions without calling a model. Housing and Procurement also report separately verified headline measures computed outside that receipt certificate; their measurement paths are documented in Appendix H.3. Replay establishes consistency with recorded execution. Scientific qualification depends on the checks in Section 6; none of the four current designs fully passes Gates 1–5 in the review snapshot.

4.2 Evaluation harness

Before execution, the kernel hashes a plan declaring worlds, seat assignments, agent configurations, seeds and budgets. During simultaneous play, it freezes all acting seats' observations before requesting actions and then applies their joint transition. Calls through its provider and tool interfaces are logged with typed outcomes. Finalization seals the evidence and produces a receipt binding the plan, world and scores.

A seat is played in one of three ways. Most models run through AERead's own harness: each action is a single call through a pinned provider route, with the seat's instructions and its current observation in and one action in the case's schema out, no tools, and no memory beyond what the observation shows. The route is checked against its pin, a seed is requested, and the reasoning setting and output limit are declared. Gemini 3.8 Flash, GLM 5.3 Flash and GPT-6 Luna in Housing (Gemini and Luna in Procurement), the Refund models and the TERMS-Bench agents are run this way. The two frontier agents, Opus 5.5 and Sol 6.1, run instead through their vendors' command-line tools, Claude Code and Codex: one fresh process per action at low effort, with the same instructions and schema, tools disabled or refused, and no session carried over. Neither tool was fully isolated from the operator's machine (Table H.1): Claude Code ran in safe mode, with none of the operator's instructions, memory or skills but with its working directory in context, and Codex still listed eight of the operator's own skills. Only the tool's version and binary are checked, and the tool exposes no seed, temperature or output limit. Through the harness the subject is the model; through a command-line tool it is the product, in-

cluding its own instructions and internal turns. A comparison across the two paths therefore compares agent configurations rather than models, and each result names its path. Counterparts, references and simple-rule baselines are scripted, call no model and replay exactly. A recorded declaration does not establish that a provider honoured it; Appendix H gives the controls each path applied and the versions behind each reported panel.

5. Case studies

We present four cases built by the maintainers (Table 5.1). They are worked examples of the framework rather than a fixed benchmark: each is a starting point for questions about the decision it models, and Section 7.2 lists the questions we consider open. The cases show how the counterpart changes measured value and how a shortfall splits into typed action classes; Section 5.3 runs an external benchmark on the same runtime. Housing, Procurement and Data center are described here and specified in Appendices A–D; Refund appears only in the cross-case comparison. Each case is a family rather than a fixed task set: a generator of worlds, a case-specific reference (a policy that uses only what the deciding side may know, or an explicitly labelled full-information upper bound), an admission rule that keeps worlds where the decision matters (its form varies by case; Appendix E.2), and a set of verifiers. The references differ in what a shortfall means: against a reachable policy it is value a feasible strategy would have kept; against the best policy given the decider's information it is the whole cost of the decisions; against a full-information bound it also includes the value of information no seat had (Appendix C).

Cases are chosen by the economic value of the decision, not by occupation (Li 2026): an activity qualifies when its value is decided rather than produced, a counterparty has its own interest and private information, and the result can be scored inside a world against a computed reference. The four examples sit in two of the five transaction classes introduced in §3 (business and household purchases); they were chosen to span the kinds of verifier and counterpart, not sampled from a representative distribution.

Table 5.1. The four cases. Each is scored in its own units against the reference in the last column.

Case	Real setting	What the agent decides	Who it faces	Scored against
Housing	informal rental listings	which listing to inspect, what to offer, and whether to sign when some listings are lemons	scripted landlords; other tenants take listings it leaves	a scripted tenant that uses only what the tenant knows
Procurement	wholesale sourcing	pay more for an established supplier, or test a cheaper newcomer, before a deadline	scripted suppliers	the best plan given what the buyer knows, solved exactly
Data center	AI-infrastructure contracting	which risk clauses to change, what price to name, and when to sign or walk away	the other side: a script, another model or the same model	the best agreement at both sides' true costs, which neither side knows
Refund	customer refund claims	what to ask, decide and pay on a claim under a written policy	the other parties to the claim, scripted	a hand-written correct resolution

How much room each case leaves is reported as a share of that case's own reference value: Tables 5.2 to 5.4 give each subject's result and gap that way, and Figure 5.2 splits the gap by action class in the case's units. The cases use four of the kinds of check defined with the runner's measurement layer (§4): an answer key, declared rules, the value of the best policy or a bound, and relative standing against another agent; none uses a rubric. Every score is computed by code, with no model in the loop; the kernel's replay recomputes each receipt's own score from its sealed events, while the headline measures of Housing and Procurement are computed by case code outside that certificate (Appendix F.2). Table E.1 lists every check.

5.1 Results

Each case's gate status (§6) bounds what its results can claim. Housing v15 passes Gates 1, 2, 3 and 5 and is partial at Gate 4. The first Data-center confirmatory is partial at every gate, chiefly because neither seat has much to infer (knowing the other side's type is worth less than 1% of the value available in a deal to the best informed policy); the second's gate review is pending. The Procurement pack reported here passes Gate 0 and is partial at Gates 1 to 5, chiefly because its environment's registration with the runtime awaits review and its code is not yet on the main branch; the earlier, saturated Procurement pack fails Gates 1 and 3. Refund V2.5 fails Gate 3. "Declared" below describes how a comparison was run, with its analysis committed before any cell, not the case: the results compare the named subjects on these worlds and are not qualified benchmark scores.

Housing. The structure follows informal rental markets on social platforms such as RedNote: listings are posted with photos a tenant cannot check before viewing, a viewing costs time, and good listings are often taken while the tenant decides. Rents, the lemon share and the inspection cost are set by the authors, not drawn from platform data. A tenant chooses among four listings, two of which are lemons. It knows its own value v_ℓ for each listing if sound, but not which two are lemons; each of the six pairs is equally likely. In each of three rounds it may pay \$25 to inspect one listing, make one offer, and sign or walk from a lease it is offered, while each listing it did not bid on is taken by outside demand with probability ½. Signing listing ℓ at rent x after k inspections pays $v_\ell - 1{,}000\cdot\mathbb{1}[\ell \text{ is a lemon}] - x - 25k$; signing nothing pays $-25k$. The hard part is information. A listing worth \$400 more than its ask if sound is worth $400 - ½\cdot1{,}000 = -\$100$ if signed blind at the ask, but $½\cdot400 - 25 = +\$175$ if inspected first and signed only when sound; inspecting a second listing takes another round, in which each listing not bid on may be taken. A landlord's reply can reveal a lemon at no cost: every world is played against one landlord kind whose reply to a low offer reveals a lemon and one whose reply does not. A blind signing is scored at its expected value given what the tenant knew, so the score follows the decision rather than the draw, with the odds conditioned on every informative landlord reply (the corrected measure, introduced in this paper; v14's declared measure conditioned on only one kind, Appendix E.4). The comparison reported here, v15, was declared before any of its artifacts and changes four things from v14: each admitted world is screened so that the best informed tenant beats a simple inspect-then-sign rule by at least \$25 (119 of 437 candidate main worlds failed), the tenant's brief states how landlords price, the corrected measure is primary from the start, and each receipt scores the deciding tenant. Its holdout of 40 worlds changes one mechanism, outside demand at 0.3 a round, and its brief says so.

Table 5.2. Housing v15, main pack of 260 worlds (200 with the favourite a lemon, 60 sound): seat 0's net under the corrected measure as a share of what the best informed tenant earns in the same worlds (\$307 a world on the main pack), mean of the two landlord arms; differences in percentage points of that value, with 95% intervals over worlds. Declared before any v15 artifact; dollar values are in the results note, and the v14 comparison is in Table E.13.

	Gemini 3.8 Flash	GPT-6 Luna
Share of the best informed tenant's value	78.7%	62.9%
Favourite is a lemon / sound	68.8% / 100.0%	46.8% / 97.3%
Minus `inspect_then_sign` (the declared beatability baseline), points	+29.3 [+24.7, +33.8]	+13.4 [+8.2, +18.6]
Minus `inspect_lowball` (v14's reachable reference), points	−16.6 [−20.8, −12.5]	−32.5 [−37.8, −27.3]
Holdout (40 worlds, outside demand at 0.3)	79.3%	61.1%

Against the best informed tenant's value, the full-information ceiling is 121%, `inspect_lowball` earns 95% and `inspect_then_sign` 49%. Under the analysis declared before any cell, Gemini leads Luna by 15.9 points [11.3, 20.4] of that value: found, but not shown to exceed the declared \$50 band (16.3 points). The A/A arm, Luna run twice under two names with different seeds, differs by −1.6 points [−6.1, +2.9], inside the band, so the band is calibrated. Both models beat the declared baseline and both stay below v14's reachable rule and the

best informed tenant. Almost the whole shortfall is in worlds where the favourite listing is a lemon: there Gemini keeps 69% and Luna 47% of what the best informed tenant earns, against 100% and 97% where the favourite is sound. In those worlds Luna signs blind twice as often as Gemini (230 against 114 blind signings in 600 cells). In the holdout Gemini leads by 18.2 points [5.8, 30.6]. All 1,720 cells completed and were verified by replay ($8.64). v15 differs from v14 in four ways at once (the screened pack, the stated pricing rule, the corrected measure as primary, and seat-0 receipts), so the levels are not comparable with v14's and no change between them is attributed to the models or to any one difference. GLM 5.3 Flash left at admission: its provider now serves another quantization than the one v14 pinned. A declared addendum ran it on another provider serving v14's quantization (600 of 600 cells, verified by replay), reported beside v15 and outside its declared contrasts. On the main pack GLM keeps 53.9% of the best informed tenant's value and does not beat `inspect_then_sign` at the declared 98.3% level (+4.5 points [−4.4, +13.3]); descriptively it trails Luna by 8.9 points [1.6, 16.2] and Gemini by 24.8 [19.0, 30.5].

Procurement. The structure follows wholesale sourcing on B2B platforms such as Alibaba's 1688. A displayed price is not a quote, a listing's quality is known only after it is checked, and new sellers of one kind often buy from the same factory. A buyer owes 50 kits by a deadline and awards one supplier: a recorded supplier with a long clean record, or one of three newcomers. Each newcomer is good, late or poor. The buyer knows only the odds, read from the record of its seller kind, and that newcomers of one kind share a source with a stated probability (0.5 to 0.95), in which case they are all of one type. Two features follow our own sourcing on 1688: a newcomer's price relative to the recorded supplier's is drawn from the price gaps in that record, so a newcomer is dearer about half the time, and a newcomer answers a quote request only a third of the time. The case's other quantities are set by the authors. Among up to ten actions the buyer may request a quote ($2 and a day; an answered quote exposes a late supplier), request a sample ($20 to $240 and three days; it always arrives and exposes a poor supplier), look up a listing's registered factory ($1; two lookups in one kind show whether it shares a source), award, or defer. An award that delivers k kits on time earns $37k - 12(50 - k)$ less the order's cost and the information bought; one that delivers fewer than 40, as a late or poor supplier does, is worth − $600 less the information bought (Appendix A.2).

The hard part is a test worth taking only for what it reveals about another supplier. Every world in the pack is one where a rule that plans a single newcomer's test exactly (what to request first, how often to repeat an unanswered quote, when to give up) pays the premium, while the best policy tests a newcomer first and beats paying the premium by at least 10%. In one world the recorded supplier asks $24.95. A newcomer asks $12.47, but its sample costs $200 and its odds are 0.20 good, 0.15 late and 0.65 poor, so testing it on its own loses money. A second newcomer of the same seller kind asks $26.87, more than the recorded supplier, so it is never worth buying; but its sample costs $30, and the two share a source with probability 0.95. Once the cheap newcomer's quote has come back clean, a sound sample of the dear one raises the cheap newcomer's odds of being good to 0.94, and a poor one lowers them to 0.01. The best policy quotes the cheap newcomer; if the quote comes back clean, it samples the dear one, looks both up, and only then pays for the $200 sample. It earns 11.8% more than paying the premium. Every action is charged what it gave up against the exact best informed policy at that point.

Table 5.3. Procurement, the 20 worlds of the declared test, two plays each: the share of the gap between paying the premium and the best policy each subject closes (a mean over worlds, each world the mean of its plays, with 95% t intervals), the loss per world below the best policy, and how many plays touched a newcomer. The worlds were drawn, and the test on Sol declared, before any play. In every one of them a rule that plans one newcomer's test exactly pays the premium and closes none of the gap.

	Share of the gap closed	Loss per world	Plays touching a newcomer
Sol 6.1 (Codex)	35% [18, 52], p = 0.0003	$32.51	26 of 40
Opus 5.5 (Claude Code)	0% in every world	$51.25	0 of 40
Gemini 3.8 Flash	0% in every world	$51.25	0 of 40
GPT-6 Luna	−14% [−21, −7]	$58.10	0 of 40
Exact rules on the same worlds			
Paying the premium; a rule that plans one newcomer's test	0%	$51.25	
Shortcut: sample the newcomer with the cheapest sample first, ignoring seller kinds	33%	$33.46	
A rule that may test two newcomers	82%	$8.38	

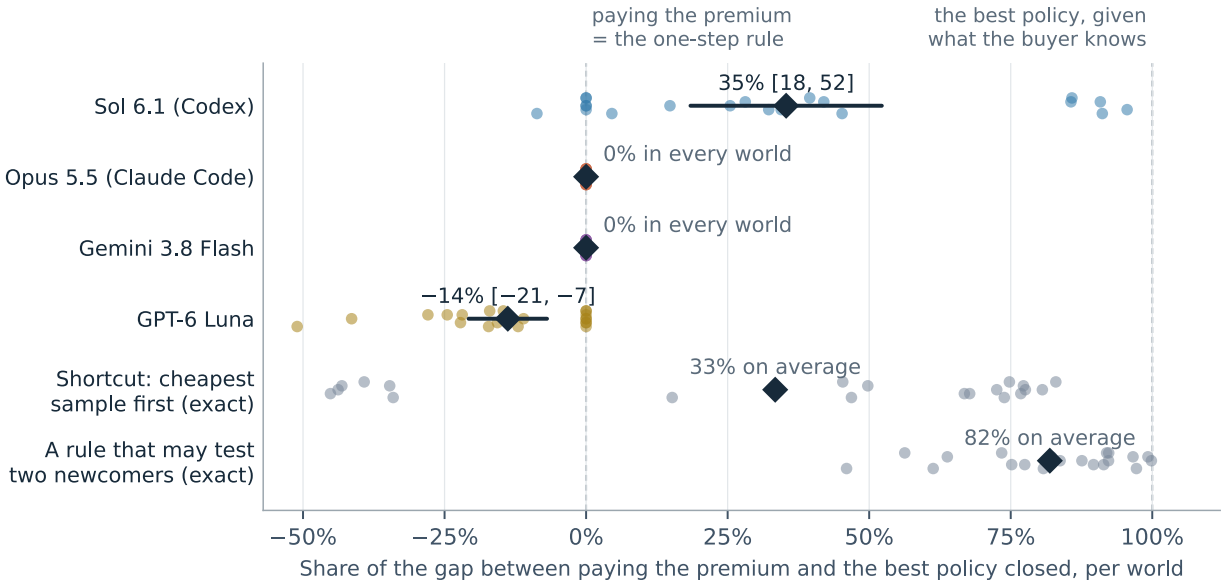


Figure 5.1. Procurement, the 20 declared worlds: the share of the gap between paying the premium and the best policy closed, world by world. Each dot is one world, the mean of its two plays; diamonds are means and bars are 95% t intervals. Paying the premium, and a rule that plans one newcomer's test exactly, sit at 0% in every world. Opus and Gemini close none of the gap in any world; Luna's orders of 50 units where 55 was best leave it below paying the premium. The gray rows are exact rules: the shortcut that ignores seller kinds and samples the newcomer whose sample is cheapest first, and a rule allowed to test two newcomers.

Sol 6.1 closed 35% of the gap (95% t interval 18–52%, p = 0.0003), where a rule that plans one newcomer's test exactly closes none. This was the one test we declared before any play, and the claim holds; it concerns Sol alone. Its play splits by world (Figure 5.1): it closes most of the gap in some worlds and none in others. The other three models never touched a newcomer. Opus 5.5 and Gemini 3.8 Flash paid the premium in every play; GPT-6 Luna did too, but ordered 50 units where 55 was best in 15 of its 40 plays, which leaves it below paying the premium. A shortcut that ignores seller kinds and samples the newcomer whose sample is cheapest first closes 33%, about Sol's share, but Sol does not play it: it opens with the shortcut's move in 2 of 40 plays, and

its tests follow price. The moves the case is built around are rare in its play: a second newcomer in 7 of 40 plays, a source lookup in 4. A rule allowed to test two newcomers closes 82%, so the headroom lies in planning one test beyond the one in hand.

Further runs on this pack are in Appendix E.4 (Table E.15): a development pilot, the declared worlds at the earlier pack's more favourable odds, and a brief corrected for four wording defects. None changes this picture. One bears on why Opus never tests: in a check with no play, it was asked how likely a newcomer is to be sound once another newcomer of its kind has passed or failed a sample, with and without a clean quote of its own first, and answered all 18 questions within 0.05 of the exact value, as Sol did. Opus reads the sharing rule and does not plan with it. These runs used the case's environment from a development branch whose registration with the runtime still awaits review (Appendix F.1). An earlier pack, in which quotes always arrive and newcomers never share a source, is saturated: a rule that looks one test ahead comes within 0.01% of the best policy (Appendix E.4, Table E.14).

Data center. The structure follows a real AI-infrastructure engagement in which a system integrator bundles hardware, data-center space, staffing and software into a working system for a client. Hardware can arrive late or defective and the site may not be ready, and the contract's warranty, delay-damages, liability-cap, site-readiness, deposit and burn-in clauses decide who bears each cost. Costs, probabilities and prices are set by the authors. Client and integrator negotiate one agreement: a position on each of the eight clauses (600 distinct agreements) and an all-in price p. Each side knows its own cost of carrying risk and only the odds of the other's: the integrator's pre-staging cost and risk charge, the client's risk charge. Over six alternating moves a side may redline any clauses and name a price, sign the standing version, or walk, and each redline after the first move costs $25k. A signed agreement c pays the integrator $p - C_I(c) - 500$ and the client $O_C - p - C_C(c)$, in $ thousands and less redline fees, where 500 is the integrator's outside option, O_C the client's, and C_I and C_C each side's expected cost of the risks the clauses leave with it, marked up by its risk charge; no deal pays only the fees. The hard part is that price only divides the value while the clauses decide how much there is, and the best clauses depend on what the other side hides. Burn-in, for example, costs the integrator $40k and, in one world, the client a week's delay worth $110k, and halves the chance of a $3 million incident the client carries; it lowers joint cost by $41k for a client with the highest risk charge and raises it by $19k or $64k for the other two, and the integrator cannot tell which client it faces. Two comparisons were declared before their runs. The first (v1) used 60 admitted worlds, 45 with a mutually beneficial deal. The second (v2) used 80 new admitted worlds, 72 with a deal. It isolated both command-line tools, stated the client's walk-away point in its brief, and admitted only worlds where the integrator has room over a price-only rule. Table 5.4 reports both; development results remain in Tables E.3 and E.7.

Table 5.4. Data-center agreement confirmatories: Sol 6.1 (Codex) and Opus 5.5 (Claude Code). Differences in a seat's earnings above its outside option, at true costs, on the deal worlds, in percentage points of the value available in a deal (the joint surplus of the best agreement: $322k a world in v1, $309k in v2); intervals are 95%, with Holm correction across each campaign's primaries. v1 has eight documented departures and a partial Gate 5 verdict (Appendix F.5); v2's gate review is pending.

Primary contrast	Worlds	Estimate [95% interval]	Holm p
v1, 14 October: the two tools not equally isolated			
Head to head: Sol minus Opus earnings, seat-averaged	45 deal	+59.1 [46.0, 72.1]	3.2e-11
Firm population: Sol minus Opus earnings, seat-averaged	45 deal	+8.2 [3.3, 13.2]	0.0017
Client signs below its walk-away point: Opus / Sol	all 60	45% vs 5% (27 and 3 of 60)	2.4e-7
v2, 19 October: both tools isolated, the walk-away point stated			

Primary contrast	Worlds	Estimate [95% interval]	Holm p
Head to head: Sol minus Opus earnings, seat-averaged	72 deal	+41.7 [31.3, 52.0]	6.1e-11
Integrator against the firm client: Sol minus Opus	72 deal	+7.5 [1.7, 13.2]	0.022
Sol as integrator minus the price-only rule	72 deal	+7.3 [2.6, 12.0]	0.008
Opus as integrator minus the price-only rule	72 deal	−0.2 [−5.5, 5.2]	0.95

In v1, Sol's lead against the firm population is detectable but smaller than the plan's minimum of 15.5 points ($50k; the 98.3% interval is [2.1, 14.3]). Head to head the opponent changes with the focal model, while the firm-population contrast holds the counterpart policy fixed, which supports a counterpart-specific reading of the larger gap. Across v1's pilot and three arms, 1,175 of 1,176 negotiations are valid ($74.41 list price). The two tools were not equally isolated in v1: Codex's context listed eight of the operator's own skills, two described in benchmark and evaluation terms, and Claude Code's listed none (Table H.1). Every v1 Sol-versus-Opus contrast carries that difference.

v2 removes it: every freeze and run probe was clean, and all 640 negotiations are valid and replayed ($55.41 list price). Its head-to-head gap is 41.7 points and its firm-client gap 7.5; each is found by Holm, and neither is shown to reach its declared minimum (32.4 and 12.9 points, $100k and $40k). Sol beats the price-only rule as integrator, and Opus earns what the rule earns. With the walk-away point stated, neither model signs below it as client (0 of 80 for either, against Opus's 27 of 60 in v1), and the client seat separates nothing (+0.8 points [−3.2, 4.9]).

The best informed policy turn by turn was computed after both runs, against the same scripted counterpart (Appendix C.3). On v2's deal worlds it gives the integrator 63.5% of the value available at the true client type, where Sol earned 33.8% and Opus 26.3%, and the price-only rule 26.5%. Sol therefore captures about 20% of the room between the rule and that reference. In both seats and both campaigns, knowing the other side's type is worth less than 1% of the value available to this policy: the case measures bargaining against a fixed counterpart far more than inference about it.

5.2 Where the shortfall comes from

Figure 5.2 separates each case's shortfall into observed action classes. The components sum to the total under a stated priority order; they describe what happened, not its cause. Each panel uses its own reference and units. Appendix E.5 gives the rules and full subject tables; E.7 gives the current Data-center breakdown and uncertainty. Table E.10 retains the exact panel values.

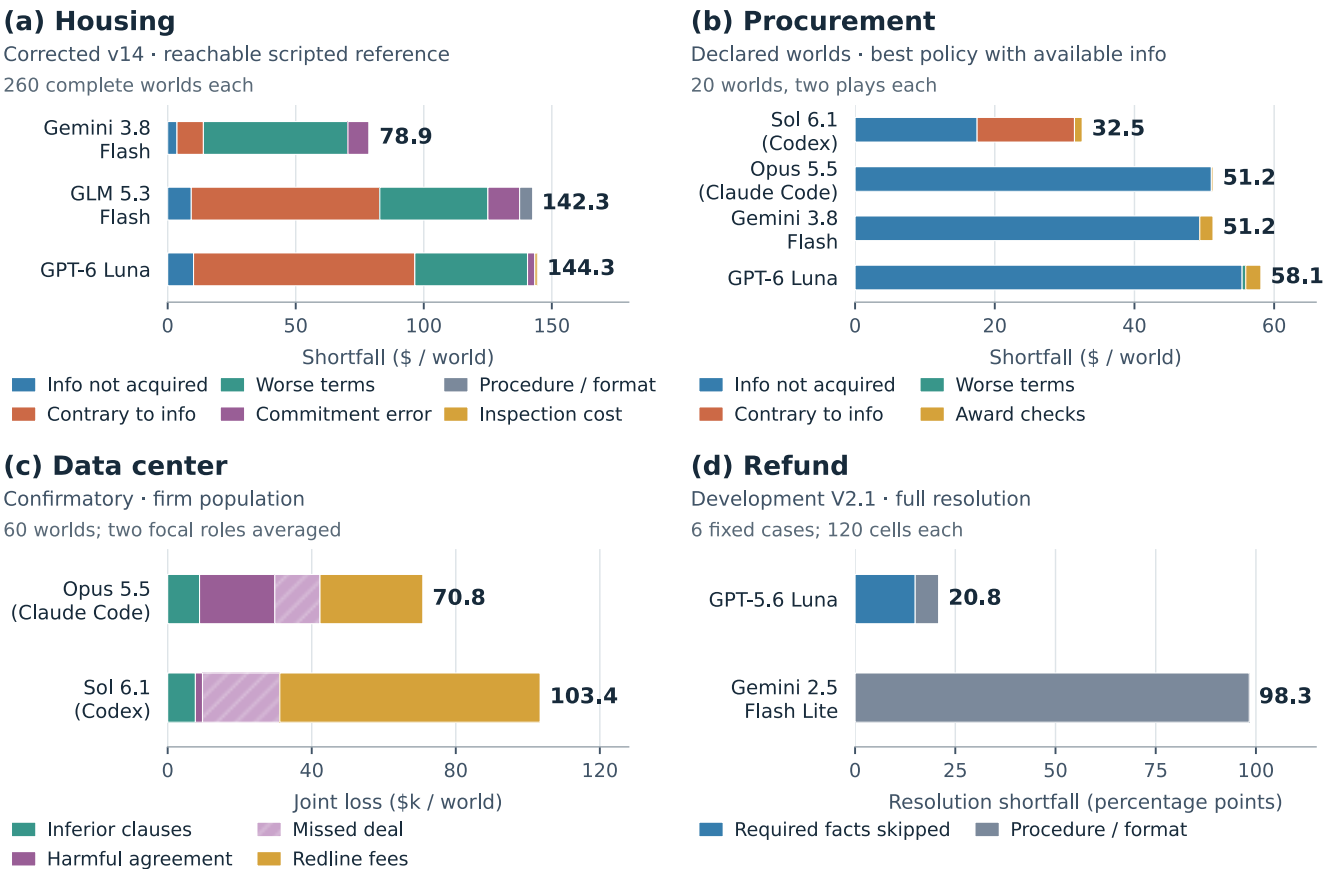


Figure 5.2. Shortfall to each case's reference. Segments account for observed actions; labels give mean totals (lower is better within a panel). Units and scales differ. Housing shows v14's three models on 260 worlds (v15 is in Table 5.2); GLM's inspection cost of −0.2 a world is in Table E.10 and is not drawn. Procurement shows all four subjects on the 20 declared worlds of Table 5.3; full subject tables are in Appendix E.5. Data center reports v1's joint loss against the full-information bound with firm counterparts at $25k per paid redline; individual below-walk-away signings are separate (Table 5.4). In Procurement, $6.33 a world of Luna's information-not-acquired segment is the worse-quantity part of awards made before a worthwhile test; the declared rules file such an award's whole loss there (Table E.10). Intervals and exact values appear in Tables 5.3, E.9 and E.10.

The labels name what happened; the mechanism behind each large segment is as follows.

Procurement: three models never test, and Sol secures the recorded supplier first. Opus, Gemini and Luna asked no newcomer for anything in any of their 120 plays: they quoted, sampled and awarded the recorded supplier, so nearly all their loss is information not acquired. Luna's loss beyond paying the premium comes from ordering 50 units where 55 was best in 15 plays; the declared classes file most of it under information not acquired (Table E.10). Sol's loss splits. Where it never tested a newcomer that the best policy would have tested, it lost $17.5 a world. Where it did test, it often first quoted and sampled the recorded supplier, spending days and actions before the newcomer test the stated odds and clock made best ($8.9 a world), or awarded while a newcomer test it had begun was still worth finishing ($3.2).

Housing (v14), Gemini: it signs leases it understood, on worse terms. 72% of its shortfall against the reachable rule comes from leases whose quality it had settled, by inspecting or because the landlord's reply proved the listing sound. It then signed on terms worse than the rule's in three ways. It chose a listing worth less than the rule's choice even at the landlord's floor (52% of this class). It paid rent above the landlord's floor (31%): a landlord accepts any offer at or above its floor, so an opening offer near the ask is accepted as made, whereas the rule opens at $0, which always draws a counter at the floor. And it inspected more than the rule (17%). It seldom signs blind or against what it knows.

Housing (v14), Luna: it signs a lemon that the landlord's reply had identified. This is 60% of its shortfall. Under the leaky landlord kind a lemon costs its landlord $1,000 less, so its floor sits about $1,000 below a sound listing's. An offer between the two floors is countered by a sound landlord at its floor but accepted by a lemon's landlord, so an acceptance there marks the listing a lemon. In Appendix A.1's example an offer of $2,000 against a sound floor of $2,375 is accepted, and signing it pays the lemon's value less the rent, −$400. Luna signed such a hold in 56 of its 520 cells, losing about $800 a cell against the rule. The rule inspects first and signs a lemon only at a rent its lemon value covers (+$225 in the example). The v14 brief did not state how landlords price, so these signings show that Luna did not act on what the acceptance implied, not that it knew what the acceptance implied; v15's brief states the rule (Table 5.2).

Refund, Gemini 2.5 Flash Lite: the case ends before it decides. In 81% of its cells it asked again for a fact it had already been given; the environment rejects a repeated request and the case ends with no decision (80.8 of its 98.3 points). In most of the rest it denied correctly but wrote the refund method as null where the policy's format requires "none" (16.7 points). It reached the answer key's outcome in 18% of cells and a wrong outcome in none.

Data center: fees, then harmful agreements. Every redline after the opening move costs $25k. The full-information reference signs the best agreement at once, so each fee counts as loss. Fees are 69.9% of Sol's and 40.3% of Opus's joint loss against the firm population across all 60 worlds. Some of those redlines buy better clauses, and a seat that does not know the other side's costs may need them, so their share alone does not show wasted negotiation. Opus's next largest part (29.5%) is harmful agreements. In 15 of the 60 worlds no agreement beats both sides' outside options, so the right move is to walk away, and Opus signed anyway.

Exploratory replay with actions fixed separates scoring sensitivity from behavioral adaptation (Appendix E.9). On the 45 deal-feasible worlds, Opus retains lower joint loss at zero fees; Sol's earnings lead rises from $26.4k to $54.8k and reverses near $48.3k per redline. Across all 60 worlds, however, removing fees reverses the joint-loss ranking: Sol loses $31.1k versus Opus's $42.3k. Fee robustness therefore depends on the measure and the worlds included.

Across the cases. Four things hold. The gap sits in a stratum the design fixes in advance, so every case reports by stratum and pools last. The counterpart is part of the measurement, so a claim names the counterparts it covers. What the brief states moves the score as much as the subject does (stating two rules cut procurement loss by $11 to $14; stating a walk-away cost cut losing signatures from 15 of 60 to 1 of 60), so the brief is a versioned part of the case. And a pattern found by looking back is a hypothesis until a declared test confirms it.

Housing, Procurement and Data center take their decision structure from real settings (Section 5.1): informal rentals, wholesale sourcing and AI-infrastructure contracting. Their parameters are set by the authors, not calibrated to field data. The experiments use controlled scenarios and a small set of agent configurations, with no human policy baseline; they do not establish general model rankings.

5.3 An external benchmark on the same runtime

TERMS-Bench evaluates bilateral price negotiation against counterparts whose type is hidden (Zhang et al. 2026). We reimplemented, from the paper's description, its alternating-offer environment, three of its six counterpart families (Candid, Taciturn and Expressive) and four of its measures as one AERead family. The counterpart runs as a scripted seat, and its random draws are sealed each turn, so replay recomputes the counterpart's moves as well as the scores. Not implemented: the Oracle-Cue reference and the gap measures built on it, the urgency regime, type-belief error and the other three families; counterpart messages are templates. The corpus has 30 cases, 15 with a zone of agreement and 15 without, each played once.

Table 5.5. TERMS-Bench reimplementation: two of the paper's agents on the same 30 cases, scripted counterpart and settings. Agreement and surplus are over the cases with a zone of agreement: 15 for GPT-4o-mini and 12 for GLM-5.1, whose three malformed moves removed those episodes. Paper values: surplus efficiency from its Table 16 (overlap regime, mean of the three families played here); agreement and conditional surplus efficiency from its Table 2, which pools six families and three regimes.

	GPT-4o-mini	GLM-5.1	Paper: GPT-4o-mini	Paper: GLM-5.1
Agreement rate where a deal exists (AGR+)	0.80	1.00	0.522	0.951
Surplus efficiency (SE+)	0.035	0.458	0.196	0.668
Conditional surplus efficiency (CSE+)	0.044	0.458	0.363	0.721

The reimplementation orders the two agents as the paper does on all three measures, but absolute levels are lower for both, and for a third model. A review of the port after these runs traced the shift to four departures from the paper's protocol, not to the agent, whose interface in the paper is also one stateless call per round. Our prompt does not state the agent's payoff or the rule against accepting above its reservation value; its observation differs from the paper's; a malformed move removes the episode where the paper applies a fallback, under which GLM-5.1's surplus efficiency would be 0.520 over all 15 cases; and our price range and zone widths are wider than the paper's. A declared reproduction will follow those fixes. The paper reports critical violations by GLM-5.1 in 5% of cases without a zone of agreement for these three families; at that rate 15 such cases would show none about half the time, and ours showed none. The port also exposed a harness effect. On one provider route, any declared reasoning budget suppressed deliberation, and GLM 5.3 Flash breached its own reservation value in 13 of 30 cases against the paper's band of 0 to 2.06%. With no reasoning declaration it breached in 2 of 30 and its surplus efficiency rose from 0.40 to 0.53; the comparison above uses that setting. This is why the reasoning condition is part of each run's declared configuration. The runtime support for a scripted seat that this family uses is still in development.

6. Quality control

The defects that mattered in the four cases were properties of the distribution, the brief, the counterpart or the analysis, and every world was individually correct when they were found: a seat-number tie-break that decided who signed the good listing, a prompt that pointed every subject at the right supplier in half the worlds and the wrong one in the other half, a hidden type tied to the pack cell so that comparisons by type compared worlds. Reading one world at a time passes all three. Quality control is therefore organised around populations of worlds, declared analyses and recorded failures. Appendix F describes each check that was applied to these cases, and only those.

Gates. A case is qualified in six gates recorded in its family's QC profile: a profile exists (0); the worlds are valid, distinct, split and leave headroom (1); actions change the intended state and every score is rebuilt from sealed evidence (2); a better-informed policy does better and shortcuts lose (3); the result supports the attribution it claims (4); and the comparison was sized, frozen and executed without tuning on outcomes (5). Each gate is typed `passed`, `failed`, `partial`, `not_run` or `not_applicable`, and only `passed` promotes. The profiles followed the initial runs; the agreement confirmatory followed its profile. None passes Gates 1 to 5 yet; Housing v15 passes all but Gate 4 (Appendix F.1).

The reference is played on every world. For each case a test plays the reference through the real environment and verifier on every admitted world, with no model, and checks that it reaches its declared result: the Housing reference and a full-information tenant reproduce the published values in all 600 v14 cells, and the full-information tenant reaches the ceiling on all 320 v15 worlds; on Procurement's harder pack the best policy and the one-step rule equal their averages over 400 plays through the environment, within four standard er-

rors, on each of the six pilot worlds (the 20 confirmatory worlds rest on the same reference and generator but were not played this way), and on the earlier pack the best policy equals the environment's average over its 27 type profiles in all 24 worlds; the data-center oracle pair loses nothing in all 116 worlds of both development packs and all 60 of the admitted confirmatory pack; Refund V2.5's release requires full credit on every case. The initial tests followed development; the admitted-pack dry run preceded the confirmatory; two of them (the data-center grade and the Housing ceiling) are shown to fail when the grade or the reference is wrong.

Other checks. Every committed pack is regenerated byte for byte from its seeds; the Housing v14 and v15 receipts replay from their sealed events; trivial agents are reported beside the subjects, Procurement's fixed rules are scored exactly, and the briefs were audited for instructions that decide the outcome. Uncertainty is computed over worlds, claims name their counterparts, and five reported analyses were declared before their runs: Housing v14 and v15, the data-center signalling test and the two agreement confirmatories.

What it found. Discovered failures are recorded in an incident log in design, operational, tooling and judgment rows that are never deleted (51, 57, 90 and 23 distinct entries across the project's logs). Of 25 defects selected for Table F.4, 14 were invisible in any single world; the 11 visible ones were found by reading code or sealed bytes, by tracing a case, or in a single gate cell (Table F.4). Nine were open when the table was compiled; a later reanalysis from the sealed actions resolves one of them for the reported panels, leaving eight open. The same checks corrected three of our own statements while this paper was written.

---

7. Discussion and conclusion

7.1 What the cases show

AERead evaluates an agent’s economic decisions by the value it obtains for a principal, given the information, resources and authority of its seat. Each score is measured against a reference whose access to information is stated, whether a feasible policy, the best policy under the seat’s information or a full-information bound, and it keeps its own units and names the counterparts it was measured against, so that every shortfall has a defined interpretation. The worked examples apply this to information acquisition, bargaining and commitment in housing, procurement and contracting, and a reimplementation of part of TERMS-Bench runs on the same runtime.

The cases teach three lessons that bear on any evaluation of this kind. The counterpart is part of the measurement: the same two agents differ by far more when they negotiate with each other than when each faces a fixed counterpart. The brief is part of the case: stating a rule it had left implicit moved scores as much as changing the agent did. And the defects that matter are often properties of the population of worlds, the brief or the analysis rather than of any single world, which is why construction and quality control belong in the method; the checks of Section 6 changed the interpretation of our own results. Loss categories locate a shortfall but do not explain it, and explaining it requires controlled follow-up comparisons. Replay establishes consistency with recorded play; construct validity requires evidence about the worlds, the information and the controls.

These properties matter most where agents are deployed. Organizations that pilot agents often stop because the business value is unclear and the risks are hard to control (Gartner 2025); a value score against a stated reference and a typed account of losses address both. Small firms face a different problem: short of staff (National Federation of Independent Business 2026), they often leave decisions like those our cases model to one person, without specialized expertise and with no review before a commitment. Delegating such a decision to an agent is attractive there, and trusting it requires evidence that the agent keeps value against the counterparts it will meet.

The present evidence has clear limits. The cases take their decision structure from informal rentals, wholesale sourcing and AI-infrastructure contracting, but their parameters are set by the authors; calibration to field data and transfer to field settings remain to be established. Coverage is limited to two transaction classes and a small set of agent configurations, with no human policy baseline. Housing v15 passes four of the five qualification gates and no case passes all five: several checks on the other cases were added after their runs, and hosted-model and CLI behavior limits attribution from declared settings. Versioned reanalyses now regenerate the corrected Housing measures and Procurement decision losses from sealed records, which strengthens their provenance, but full verification against original execution still requires retained logs (Appendix G), and the reanalyses neither qualify the campaigns retroactively nor resolve Housing's historical seat attribution (Appendix H). The results therefore compare named agents on stated worlds; they do not establish general model rankings or deployment readiness.

7.2 Open research directions and contributions

AERead is shared infrastructure for studying how agents make economic decisions. Its code is open source under the Apache-2.0 license (github.com/aeread-org/AERead; contact@aeread.org), and contributors can pursue their own questions with the existing cases, counterparts and evidence. The results above leave four questions open, each suited to a different kind of contribution.

- **Counterpart populations.** The Data-center gap between two agents is 42% of the value available in a deal head to head and 7% against a fixed script. How far does an agent's measured skill depend on whom it faces, can rankings reverse across counterpart populations (none did in our runs), and which populations separate distinct strategic capabilities? The scripted styles, head-to-head pairings and counterpart seats are in place.
- **Validity of populations of worlds.** Of 25 defects selected from our record, 14 were invisible in any single world. Which checks find such defects before a campaign runs, and what should qualify a case? The quality-control gates, the incident record and the failed campaigns are available for study.
- **Information and judgment.** Stating a rule the brief had left implicit moved scores as much as changing the agent. Separating what an agent was told from how it decided, and computing best-informed references for richer cases, are open methodological problems.
- **New economic decisions.** A case for another activity supplies its rules, worlds and reference; execution, replay, counterpart seats and the quality-control standard come with the runtime.

What is ready and what is not: the runtime, its sealed evidence and replay, and the case layout are in use, and Housing, Procurement and Data center each have a world generator, references and scripted counterparts. No case is yet fully qualified: Housing v15 passes four of the five gates. Section 6 gives every gate verdict and Appendix F each campaign's open checks, so a contributor can see what a case still needs before its results can rank agents.

The next steps follow from the limits in Section 7.1. Future campaigns should bind actual seat assignments and headline scorers into frozen evaluation identities and run the required controls before any comparison, and the TERMS-Bench reimplementation needs the protocol fixes described in Section 5.3 before a declared reproduction. Practitioner review would calibrate the cases' parameters, human baselines would show where agents stand against people, and externally contributed cases would test whether the structure serves authors other than its maintainers. AERead's present contribution is that structure, a common way to construct evaluations of economic decisions, together with an evidence-based account of the checks needed to interpret what they measure.

References

Barres, Victor, Honghua Dong, Soham Ray, Xujie Si, and Karthik Narasimhan. 2025. "τ²-Bench: Evaluating Conversational Agents in a Dual-Control Environment." *arXiv Preprint arXiv:2506.07982*. https://arxiv.org/abs/2506.07982.

Bianchi, Federico, Patrick John Chia, Mert Yuksekgonul, Jacopo Tagliabue, Dan Jurafsky, and James Zou. 2024. "How Well Can LLMs Negotiate? NegotiationArena Platform and Analysis." *arXiv Preprint arXiv:2402.05863*. https://arxiv.org/abs/2402.05863.

Feng, Yeqi, Yuxin Chen, and Tianxing He. 2026. "SidConArena: An Environment Evaluating Agents in Open-Ended, Positive-Sum Bargaining Game." *arXiv Preprint arXiv:2606.27397*. https://arxiv.org/abs/2606.27397.

Fish, Sara, Julia Shephard, Minkai Li, Ran I. Shorrer, and Yannai A. Gonczarowski. 2025. "EconEvals: Benchmarks and Litmus Tests for Economic Decision-Making by LLM Agents." *arXiv Preprint arXiv:2503.18825*. https://arxiv.org/abs/2503.18825.

Gartner. 2025. "Gartner Predicts over 40% of Agentic AI Projects Will Be Canceled by End of 2027." https://www.gartner.com/en/newsroom/press-releases/2025-06-25-gartner-predicts-over-40-percent-of-agentic-ai-projects-will-be-canceled-by-end-of-2027.

He, Horace, and Thinking Machines Lab. 2025. "Defeating Nondeterminism in LLM Inference." https://thinkingmachines.ai/blog/defeating-nondeterminism-in-llm-inference/.

Lanctot, Marc, Edward Lockhart, Jean-Baptiste Lespiau, Vinicius Zambaldi, Satyaki Upadhyay, Julien Pérolat, Sriram Srinivasan, et al. 2019. "OpenSpiel: A Framework for Reinforcement Learning in Games." *arXiv Preprint arXiv:1908.09453*. https://arxiv.org/abs/1908.09453.

Li, Chenyu. 2026. "Activities, Not Occupations." https://aeread.org/blog/activities-not-occupations.html.

Liu, Xianyang, Shangding Gu, and Dawn Song. 2026. "AgenticPay: A Multi-Agent LLM Negotiation System for Buyer-Seller Transactions." *arXiv Preprint arXiv:2602.06008*. https://arxiv.org/abs/2602.06008.

Merrill, Mike A., Alexander G. Shaw, Nicholas Carlini, Boxuan Li, Harsh Raj, Ivan Bercovich, et al. 2026. "Terminal-Bench: Benchmarking Agents on Hard, Realistic Tasks in Command Line Interfaces." *arXiv Preprint arXiv:2601.11868*. https://arxiv.org/abs/2601.11868.

National Federation of Independent Business. 2026. "NFIB Jobs Report: Small Business Hiring Slows in September." https://www.nfib.com/news/press-release/nfib-jobs-report-small-business-hiring-slows-in-september/.

Patwardhan, Tejal, Rachel Dias, Elizabeth Proehl, Grace Kim, Michele Wang, Olivia Watkins, Simón Posada Fishman, et al. 2025. "GDPval: Evaluating AI Model Performance on Real-World Economically Valuable Tasks." *arXiv Preprint arXiv:2510.04374*. https://arxiv.org/abs/2510.04374.

Sun, Yiyou, Xinyang Han, Weichen Zhang, Yuanbo Pang, Tianyu Wang, Yuhan Cao, Yixiao Huang, et al. 2026. "Agents' Last Exam." *arXiv Preprint arXiv:2606.05405*. https://arxiv.org/abs/2606.05405.

Terry, J. K., Benjamin Black, Nathaniel Grammel, Mario Jayakumar, Ananth Hari, Ryan Sullivan, Luis S. Santos, et al. 2021. "PettingZoo: Gym for Multi-Agent Reinforcement Learning." In *Advances in Neural Information Processing Systems*. Vol. 34. https://arxiv.org/abs/2009.14471.

Trivedi, Harsh, Tushar Khot, Mareike Hartmann, Ruskin Manku, Vinty Dong, Edward Li, Shashank Gupta, Ashish Sabharwal, and Niranjan Balasubramanian. 2024. "AppWorld: A Controllable World of Apps and People for Benchmarking Interactive Coding Agents." *arXiv Preprint arXiv:2407.18901*. https://arxiv.org/abs/2407.18901.

Zhang, Erica, Fangzhao Zhang, Aneesh Pappu, Batu El, Jose Blanchet, Susan Athey, Jiashuo Liu, and James Zou. 2026. "TERMS-Bench: Diagnosing LLM Negotiation Agents Beyond Deal Rate." *arXiv Preprint arXiv:2605.13909*. https://arxiv.org/abs/2605.13909.

Zhu, Yuxuan et al. 2025. "Establishing Best Practices for Building Rigorous Agentic Benchmarks." https://arxiv.org/abs/2507.02825.

Appendix A. Case specifications

Following TERMS-Bench (Zhang et al. 2026), each case is written as an environment Γ (outcomes, private types and their prior, payoffs), a protocol (seats, order of play, actions, what each seat observes, termination), and fixed counterpart policies (Appendix B). Refund is not specified here. Symbols are local to each case; each case's table lists them, with the kind of quantity each is. The evaluated seat is subscript A, a counterpart B. In

controlled comparisons, the prior and counterpart policies are fixed across evaluated agents; head-to-head role swaps change the opponent as well, so they compare pairings. Agent configurations also differ in the settings recorded in Appendix H. The reference policies of Appendix C are computed in the same Γ.

Notation used across cases. $\mathbb{1}[\cdot]$ (or $\mathbb{1}(\cdot)$) is 1 when the condition inside holds and 0 otherwise. U(a, b) is a uniform draw between a and b. $\mathbb{E}$ is an expectation over the outcomes the case lists, and $\lfloor x \rfloor$ rounds x down. D is each case's loss symbol: the lemon loss ($1,000) in Housing, the value of supplying nothing (−$600) in Procurement. Each case's table defines its other symbols, and Appendices C and D point back to it when they reuse one.

A.1 Housing: lemons with outside demand (identities v14 and v15)

v15 plays the same rules. Its brief states the landlords' pricing rule, its worlds come from a screened pack (Appendix E.2), and its holdout's outside demand leaves at 0.3 a round.

Environment. A world seed w fixes four listings $\ell \in \{0, 1, 2, 3\}$ and six tenant seats. Seat 0 is the evaluated tenant; seat $i \in \{1, \ldots, 4\}$ is outside demand for listing $i - 1$; seat 5 never acts. Exactly $m = 2$ listings are lemons, $\Lambda \subset \{0, \ldots, 3\}$, drawn uniformly from a random stream separate from the one that sets values, asks and costs. Seat 0 signs at most one lease (ℓ, rent), or none. Its payoff after signing ℓ with k inspections is

$u_A = v_\ell - D \cdot \mathbb{1}[\ell \in \Lambda] - \text{rent} - c \cdot k$, and $u_A = -c \cdot k$ if it signs nothing,

where D is the lemon loss and c the inspection fee (Table A.1).

Seat 0 privately knows v_ℓ, its value for each listing if sound. Each landlord privately knows its listing's quality and cost. The prior is uniform over the six possible lemon pairs. The landlord kind is not drawn: both kinds are played on every world.

Table A.1. Housing parameters.

Symbol	Meaning	Value
T	rounds	3
m	lemons	2 of 4
D	lemon loss (cuts the tenant's value and the landlord's cost alike)	$1,000
c	inspection fee	$25
a_ℓ	ask	$\lfloor U(1500, 3200) \rfloor$
v_ℓ	seat 0's value if sound	$a_\ell + 10 \cdot s_\ell + U(-20, 20)$, with $s_\ell = 100(0.6\kappa_\ell + 0.4\iota_\ell) - 30$ and κ, $\iota \sim U(0, 1)$; κ_ℓ and ι_ℓ are listing attributes shown on the board, which move seat 0's value but say nothing about quality
c^S_ℓ	a sound listing's cost	$\max(0, a_\ell - U(20, 80))$
c^L_ℓ	a lemon's cost	$\max(0, c^S_\ell - 1000)$
μ	landlord margin: a landlord's target is its reservation plus μ, capped at the ask (B.1)	$25
β	outside-demand rate per round	½ (v15's holdout: 0.3)

On the main pack asks run from $1,501 to $3,198, and seat 0's gain $v - a$ from −$275 to +$676.

Protocol. Each round $r = 0, 1, 2$ has four phases. (1) *Inspect*: seat 0 may pay c to inspect one open listing it has not inspected; the quality appears in its observation at once. (2) *Contact*: seat 0 makes one offer (ℓ, x) with ℓ open and $x \geq 0$, or passes; offering again on a listing it walked from is legal. (3) *Respond*: each landlord ap-

plies its policy (B.1) to its inbox; each listing creates at most one binding hold. (4) *Commit*: a seat holding a hold signs or walks; a missing or invalid action is a walk, and every hold expires at the end of the phase. Signing ends seat 0's play; the market ends after round 2 or when every listing is leased.

Observation of seat 0. The board (asks, uninformative listing attributes, open or leased), its rejected listings and active hold, its values v, the lemon count and share, the loss, the fee, its own inspections and spend, and `your_history`: its own offers, each hold with the binding rent and whether the landlord accepted or countered, and each sign or walk. The sealed prompt states the payoff, that a fixed landlord "may accept or make one binding counteroffer", that "a counteroffer can itself convey information about listing quality", the departure rule, that a listing seat 0 bid on is protected that round, the three-round horizon, that walking does not close a listing, and the order of phases. It does not state the landlord's pricing rule or the cost range.

Example: a reply and a signing. A listing asks a = \$2,400; its sound cost is c^S = \$2,350, so a lemon's is c^L = \$1,350, and seat 0 values it at v = \$2,600 if sound. With the \$25 margin (B.1) a sound listing's landlord targets \$2,375, a lemon's leaky landlord \$1,375, and a lemon's pooled landlord \$2,375. Seat 0 offers \$2,000. A sound landlord counters at \$2,375; a pooled lemon landlord also counters at \$2,375; a leaky lemon landlord accepts at \$2,000, and only a lemon's landlord would, so the acceptance itself marks the lemon. Signing pays v − rent = +\$225 for the sound listing at \$2,375, v − D − rent = −\$400 for the leaky lemon at \$2,000, and −\$775 for the pooled lemon at \$2,375; inspecting first costs \$25 and shows the quality under either kind. Had seat 0 offered \$0, the leaky lemon landlord would have countered at \$1,375, which also marks the lemon, and signing there pays +\$225.

A.2 Procurement: pay or test (`pay_or_test_v1`, the earlier pack)

Environment. One buyer and one component. The suppliers are one recorded supplier E (established: it has a sales record) and three newcomers N_1–N_3. Each supplier has a type $\theta \in \{G, L, P\}$ = {(on-time 0.97, yield 0.97), (0.55, 0.97), (0.97, 0.55)}; E is always G. Newcomer types are independent with prior $\pi_i = (\gamma_i, \lambda_i, \rho_i)$ = $((40 - a_i - b_i)/40, a_i/40, b_i/40)$, read from the cohort record of N_i's seller kind (a_i of 40 tracked sellers unreliable on delivery, b_i failed inspection). That gives $3^3 = 27$ type profiles, each with positive weight; γ ranges from 0.2 to 0.7.

Table A.2. Procurement parameters.

Quantity	Value
Kits owed / minimum service	50 / 40 (an award completing fewer than 40 is void)
Revenue per kit / shortfall penalty	\$37 / \$12 per kit short
Supplying nothing, D	−\$600 (= −12·50), plus any information already spent
Cash budget	\$2,500 (never binds)
Deadline T	10–12 days (clock "none"), 13–14 (tight), 17–18 (loose)
Action budget	10
Order sizes	50, 55 or 60 units (minimum order quantity, MOQ, 50; step 5; capacity 60); shipping \$0.60 a unit
E	price U[24, 28] rounded to \$0.05; lead 6 days; sample \$30 and 2 days; a record of 48–96 orders, 3% late, 3% defective
Newcomer i	price $p_E(1 - \delta_i)$, δ drawn without replacement from {3, 5, 7, 9, 12, 15, 18, 21}%; lead 6–8 days; sample \$30–80 in \$10 steps and 3 days
Cohort record	$a_i \in \{4, 8\}$, $b_i \in \{8, 14, 20, 24\}$ of 40

Payoff. For an award line of q units at price p, yield y, on-time rate r and lead ℓ, made on day t with information spend I: units = q·y·r if t + ℓ ≤ T and 0 otherwise; kits k = min(⌊units⌋, 50); recovery R = 0.9·q(1 − y)·p; cost C = qp + 0.6q + qp·0.12·15/365 − R + R·0.12·10/365 + I; and the margin is M = 37k − 12(50 − k) − C if k ≥ 40 and the line is valid, otherwise D − I. In C, 0.6 is the shipping fee per unit; the order is financed at 12% a year for 15 days (the 45-day working-capital horizon less the 30-day payment terms); and a defect refund arrives 10 days after the claim, financed at the same rate. For a good supplier q = 55 is optimal at every pack price and gives A(p) = 1817 − 53.79p before information cost: 55 units at y = r = 0.97 complete all 50 kits, so 1817 = 37·50 − 0.6·55 and 53.79 = 55(1 + 0.12·15/365) − 1.485(1 − 0.12·10/365), where 1.485p = 0.9·55·0.03·p is the recovery.

Protocol. One action at a time until a terminal action, the action budget, or an invalid action; the last three are worth D − I.

Table A.3. Procurement actions.

Action	Cost	Reveals
request a quote	$2, 1 day	a binding offer at the listed terms and the true on-time rate, which identifies L
request a sample	the listed fee and days	100 inspected units, yield 0.97 or 0.55, which identifies P; no noise
inquire	$1, 1 day	verbal claims; every seller claims 0.97 and 0.97
counter-offer	$1, 1 day	nothing in this pack (B.2)
check an award	$0, 0 days, one action	a dry run of the award; nothing beyond what quotes and samples showed
submit an award	terminal	eligible only with an unexpired formal offer, a verified sample and the exact variant (the component version the order requires)
defer	terminal	worth D − I

Every listing shows a claimed yield of 0.97 whatever the supplier's type. The two briefs differ by two lines; the second adds that the buyer "places many orders like this one and is judged on its average margin across them", and that "a formal quote is the supplier's final price and terms. A counter_offer cannot improve them."

Example: what a quote is worth. The recorded supplier lists $26.00, so awarding it 55 units earns A(26.00) = 1817 − 53.79·26.00 = $418.46. A newcomer lists 15% less, $22.10, worth A = $628.24 if good; its cohort record shows 8 of 40 sellers late and 14 of 40 failing inspection, so γ = 0.45, λ = 0.20, ρ = 0.35, and its sample costs $50. A quote costs $2 and identifies a late supplier, which saves the $50 sample with probability λ: worth 0.20·50 − 2 = $8 before the sample. Testing the newcomer (quote, then sample if the quote is clean) instead of paying the premium is worth γ(628.24 − 418.46 + 30) − 2 − (1 − λ)·50 = $65.90 in expectation, where the $30 is the recorded supplier's sample the buyer no longer needs, provided a failed test still leaves time to return to the recorded supplier before the deadline.

A.2.1 The pack reported in Section 5.1 (`pay_or_test_v3`)

Section 5.1 reports a harder pack; the pack above (Table E.14) is its predecessor. The payoff, the three types, the kits owed, the action budget and the order sizes are unchanged. What changes:

Table A.6. What the harder Procurement pack changes.

Quantity	Earlier pack (above)	Harder pack
Newcomer price	$p_E(1 - \delta)$, δ from 3% to 21%	$p_E(1 - \delta)$, δ drawn from the price gaps in the authors' own sourcing record (−189% to +66%; a newcomer is dearer about half the time)
Shared source	none: newcomer types independent	newcomers of one seller kind share a source with probability $s \in \{0.5, 0.65, 0.8, 0.95\}$; if they do, all take one type drawn from their cohort's odds
Quote	always answered	a newcomer answers each request with probability 1/3, independently; an unanswered request costs its $2 and day and shows nothing; the recorded supplier always answers
Sample	$30 to $80, 3 days	$20 to $240, 3 days; always arrives
Source lookup	none	$1, 0 days: names the listing's registered factory; two newcomers of one kind name the same factory if and only if they share a source
Inquiry, counter-offer	offered; each can only lose	removed
Financing, cash budget, shipping	12% a year; $2,500; $0.60 a unit	none; never binds and is not shown; folded into the listed price
Cohort record	$a_i \in \{4, 8\}$, $b_i \in \{8, 14, 20, 24\}$ of 40	$a_i \in \{2, 4, 6\}$, $b_i \in \{18, 22, 26, 30\}$ of 40
Deadline	three clocks	14 to 22 days
Admission	none	a rule that plans one newcomer's test exactly loses at least $10 to the best policy and pays the premium first; every fixed rule and a kind-blind shortcut lose at least $10; the best policy beats paying the premium by at least 10% of its value; without a shared source the one-step rule is within $5 of the best policy; information is worth at least $10

The best informed policy is computed exactly by backward induction over what the buyer has seen (each supplier's test results, which listings it looked up, and whether each kind's sharing is settled), with an unanswered quote and a lookup as chance outcomes; its value equals the environment's own average over 400 plays of each pilot world within four standard errors. The 20 worlds of the declared test are the first admitted seed of each of 20 blocks from seed 9,400,000 (5,164 seeds scanned); their twins at the earlier pack's odds keep every listing, price, sample term, clock and sharing chance and redraw only the cohort records.

A.3 Data center: negotiating into an agreement

Outcomes. $X = (C \times \mathbb{R}_+) \cup \{\varnothing\}$: a clause vector c with an all-in price p, or no deal.

Table A.4. The eight clauses.

Clause	Positions
warranty W	none, fix, fix and delay damages
damages rate d	50, 100, 200 a week (effective only under fix and delay damages)
liability cap K	uncapped, 1,500, 500
site readiness R	client, integrator

Clause	Positions
consequential loss Q	excluded, included
deposit δ	none, 25%, 50% of hardware
escrow E	no, yes (effective only with a deposit)
burn-in B	no, yes

Positions without effect are normalised away, leaving $|C| = 5{\cdot}3{\cdot}2{\cdot}2{\cdot}5{\cdot}2 = 600$ distinct agreements, all reachable by redlining. The document opens on the integrator's standard draft c^0, set by its playbook (coordination, managed or turnkey), a default its private type fixes (below).

Private types. The integrator $\theta_I = (t, \rho_I)$ has a pre-staging cost $t \in \{40, 120, 200\}$ and a risk charge $\rho_I \in \{0.3, 1.2\}$, six types equally likely; the type fixes the playbook ($\rho_I \geq 1$ gives coordination, otherwise $t \leq 40$ gives turnkey, otherwise managed), so the draft narrows the client's prior and in 22 of the 58 development worlds (Appendix E.2) reveals the type. The client has a risk charge $\rho_C \in \{0.15, 0.75, 1.55\}$, declared prior one third each; in the packs ρ_C is fixed by the world's cell (one of eleven outside-option situations, Appendix E.2) rather than drawn.

Table A.5. Data-center symbols. Money in $ thousands; ranges are the generator's uniform draws on the base pack (some cells widen them), public to both sides unless marked private.

Symbol	Kind	Meaning	Value or range
a, s	decision (0/1)	the integrator pre-stages the cluster; it prepares the client's site (only if it carries site readiness)	chosen to minimise its cost
e_{def}, e_{late}, e_{inc}, e_{fail}	event indicator (0/1)	a compatibility defect; a late facility; an incident after handover; the integrator fails and the deposit is lost	independent given a, s and the clauses
q_0, q_1	probability	a defect without and with pre-staging	0.25–0.35; 0.04–0.08
u	probability	a late facility (×0.4 if the site is prepared)	0.20–0.30
ι	probability	an incident (×0.5 with burn-in)	0.04–0.06
ν	probability	the integrator fails (counts only with a deposit not in escrow)	0–0.015
F	money	fixing a defect	200–300
S	money	standby cost of a late facility	150–220
Φ	money	an incident's loss	2,500–3,500
v	money per week	the client's cost of a week's delay	90–130
w_{def}, w_{late}	duration (weeks)	delay from a defect; from a late facility	3–5; 2–4
H, σ	money	hardware; services	9,600; 600
δ	share	deposit, as a share of hardware	0, 0.25 or 0.5 (clause)
t	money, private	the integrator's pre-staging cost	40, 120 or 200
ρ_I, ρ_C	risk charge, private	each side's markup on the expected loss it carries	0.3 or 1.2; 0.15, 0.75 or 1.55
x	money	preparing the client's site	30–90
r_C	rate per year	the client's cost of capital on a deposit (13 weeks)	0.05–0.07
team	money	the client's own team if it manages the deployment	600–1,000
T	money	the rival turnkey offer	stated in both briefs
m	money	the integrator's outside option	500
O, Y	money, per outcome	what the contract says the integrator owes; what it pays after the cap K	below

Events and costs. Given a and s, the four events are independent: $P(e_{def}) = q_1$ if a else q_0; $P(e_{late}) = u \cdot (0.4$ if s else 1); $P(e_{inc}) = \iota \cdot (0.5$ if burn-in else 1); $P(e_{fail}) = \nu$ if there is a deposit and no escrow, else 0. In each of the 16 outcomes the real loss is

$$L = e_{def} \cdot (F + w_{def} \cdot v) + e_{late} \cdot (S + w_{late} \cdot v) + e_{inc} \cdot \Phi + e_{fail} \cdot \delta H,$$

the integrator owes

$$O = e_{def} \cdot [F \cdot \mathbb{1}(W \neq \text{none}) + d \cdot w_{def} \cdot \mathbb{1}(W = \text{fix and delay})] + e_{late} \cdot S \cdot \mathbb{1}(R = \text{integrator}) + e_{inc} \cdot \Phi \cdot \mathbb{1}(Q = \text{included}),$$

and pays $Y = \min(O, K)$, the cap applying to the total in each outcome. With $\mathbb{E}$ the expectation over the outcomes, the integrator's cost is

$$C_I = H + \sigma + a \cdot t + s \cdot x + 40 \cdot B + (1 + \rho_I) \cdot \mathbb{E}[Y] + \mathbb{1}(R = \text{integrator}) \cdot 1.5 \cdot P(e_{late}) \cdot S - \mathbb{1}(\text{deposit, no escrow}) \cdot 0.14 \cdot \delta H \cdot 13/52;$$

its best response (a^*, s^*) minimises C_I, and $IC(c) = C_I(a^*, s^*)$. The client's cost, excluding price, is

$$CC(c) = (1 + \rho_C) \cdot \mathbb{E}[L - Y \mid a^*, s^*] + \mathbb{1}(\text{deposit}) \cdot r_C \cdot \delta H \cdot 13/52 + 10 \cdot \mathbb{1}(\text{escrow}) + v \cdot \mathbb{1}(\text{burn-in}).$$

In these, B is 1 when the contract includes burn-in (Table A.4), whose crew costs the integrator 40; 1.5 is the integrator's surcharge on expected standby it cannot control, paid when it carries site readiness; 0.14 is its capital rate per year on a deposit it holds without escrow; 13/52 is the thirteen weeks between signing and delivery, as a share of a year; and 10 is the client's escrow fee.

Outside options and payoffs. The client's outside option is $O_C = \min(T, M)$, the cheaper of a rival turnkey offer T and managing the deployment itself, $M = H + \text{team} + (1 + \rho_C)[q_0(F + w_{def} \cdot v) + u(S + w_{late} \cdot v) + \iota\Phi]$; the integrator's is $m = 500$. With n_i side i's redlines after the first move, a signed (c, p) pays $U_I = p - IC - m - 25 \cdot n_I$ and $U_C = O_C - p - CC - 25 \cdot n_C$; no deal pays $U_i = -25 \cdot n_i$.

Protocol. Six moves, the integrator on odd moves. Move 1: redline or walk (no price is yet on the table). Moves 2–5: redline, sign the other side's standing version at its price, or walk. Move 6 (client): sign or walk. A redline sets any clauses to listed positions and states a positive all-in price, and may carry a note of up to 400 characters. A move that fails to parse or is illegal ends the episode as an invalid action; each move gets one attempt. Each side sees its own brief (every public number, its own type, the other side's possible types with their odds) and, at each move, the document with the latest tracked change on each clause, the price on the table and the full history including notes.

Example: one clause. Take $v = 110$, $\iota = 0.05$ and $\Phi = 3{,}000$ with consequential loss excluded, so the client carries incidents. Adding burn-in costs the integrator its crew, 40, and the client a week's delay, $v = 110$. It halves the incident chance, cutting the client's expected loss by $0.025 \cdot 3{,}000 = 75$ before its risk charge, which it values at $(1 + \rho_C) \cdot 75$: 86 at $\rho_C = 0.15$, 131 at 0.75 and 191 at 1.55. Joint cost therefore changes by $40 + 110 - (1 + \rho_C) \cdot 75$: +64, +19 or −41. So burn-in belongs in the best agreement only for a client with the high charge, which the integrator does not know; that is why the best agreement depends on the other side's type.

Appendix B. Counterpart policies

B.1 Housing: landlords and outside demand

Landlord. Each landlord has a reservation ρ_ℓ: $\rho_\ell = c_\ell$ (its own cost) under the *leaky* kind, and $\rho_\ell = c^S{}_\ell$ for every listing under the *pooled* kind. Its target is $\tau_\ell = \max(\rho_\ell, \min(a_\ell, \rho_\ell + \mu))$. Given the highest offer x^* in its inbox (ties to the lower seat), it accepts (hold at x^*) if $x^* \geq \tau_\ell$ and otherwise counters at τ_ℓ. A $0 offer there-

fore always draws a counter at τ_ℓ. On the main pack a sound listing's target lies in [a − 55, a]; a lemon's under the leaky kind in [a − 1055, a − 996]; a lemon's under the pooled kind equals a sound one's. A reply reveals quality only to an offer below the sound floor $\min(a_\ell, c^S_\ell + 25)$; any offer at or above it is accepted by both.

Outside demand. Listing ℓ is wanted in round r iff $H(\text{w-asks}, \ell, r)/2^{64} < \beta$, with H the first eight bytes of a SHA-256 of the world's asks, the listing and the round. A wanted open listing is taken: its outside seat offers \$0, the landlord counters at τ_ℓ, and the seat signs in the same round. If seat 0 made an offer on ℓ that round, its offer wins the tie and ℓ is not taken. The schedule depends on neither quality, nor landlord kind, nor seat 0's play; on the main pack 52.1% of listing-rounds are takes.

B.2 Procurement: suppliers

Quotes carry the true base terms, which equal the listing. Samples are exact. A counter-offer is accepted only if every term lies between the supplier's limit and the current offer, and every limit equals the quoted value (price floor = quoted price, minimum order 50, 30-day payment and refund windows), so only a restatement of the current terms is accepted: quotes are final. Awards are scored on the true private terms.

B.3 Data center: the scripted negotiator

The script plays seat σ, knows its own type exactly, and knows the other side through the declared types Θ and their prior β_0. Shares (s, s_{min}) are fractions of the value on the table Π; τ, χ and V are money; r, α and κ are dimensionless weights. For a type θ write v_σ and $v_{-\sigma}$ for the two sides' surpluses and $G(c; \theta) = v_\sigma + v_{-\sigma}$, which does not depend on price.

1. **Belief.** $\beta \propto \beta_0$ restricted to types for which every redline the other side made was worth at least its outside option to it; an update that would leave no type is skipped.
2. **Value on the table.** $\Pi = \max(0, \max \text{ over } c \text{ of } E_\beta\, G)$.
3. **Ask.** With N own proposals (3 as integrator, 2 as client) and $\Delta = (s_{open} - s_{final})/(N - 1)$, the ask share starts at $s_1 = s_{open}$ and moves to $s_{k+1} = \max(s_{final}, s_k - \Delta\kappa_k)$, with $\kappa = (1 - r) + r\cdot\min(2, \max(0, \eta/\Delta))$, where η is how far the other side moved toward the script since its last proposal, measured as a share of Π. Collapse holds s_{open} until its last proposal, then asks s_{final}.
4. **Proposal.** For each set of the most valuable remaining types, the script picks the clause set maximising expected joint value over that set (price only keeps the standing clauses), prices it at its break-even plus the ask share of that value, capped at what the least willing type in the set would accept, rounds to \$5k, and keeps the set with the largest expected value to itself, types outside the set counting as refusing.
5. **Move.** With τ the expected value to the script of the standing offer and $\chi = \alpha\cdot V - 25$ the value of proposing, it signs if $\tau \geq \max(0, s_{min}\cdot\Pi)$ and either it cannot propose or $\tau \geq \chi$; otherwise it redlines if it has a proposal with $\chi > 0$; otherwise it walks. It never reads or writes a note.

Table B.1. Counterpart styles of the data-center case. (Defaults r = 1, α = 0.8, $s_{min} = 0$.)

Style	Role	Opening ask s_{open}	Final ask s_{final}	Least share s_{min}	Reciprocity r	α	Other
firm population (`drawn`)	core	U(0.60, 0.90)	$\max(0.45, s_{open} - U(0.05, 0.30))$	U(0.10, 0.35)	0.5 or 1.0, equal odds	0.8	drawn per world and seat
soft	core	0.55	0.25	0	0	0.4	concedes on a clock
price only	adversarial	0.70	0.50	0.20	1	0.8	never changes a clause

Style	Role	Opening ask s_{open}	Final ask s_{final}	Least share s_{min}	Reciprocity r	α	Other
collapse	adversarial	0.90	0.30	0	unused	0.8	holds, then gives way on its last proposal
overreach	adversarial	1.30	0.70	0.20	0	0.8	priced past every type of the other side while $s \geq 1$
prober	adversarial (information)	as the firm population					asks in notes for costs and a confidential item; believes a stated cost matching a type

1. **Firm population.** Each world and seat draws its own style, so the counterpart is a population rather than one rule. It concedes no faster than the model does in about half of world-seats ($r = 1$) and refuses any offer below its least share. It tests whether pressing pays against a counterpart that holds. Its draw hashes the whole world, including the client's hidden charge, which leaks nothing to the model; the declared test used a fixed style.
2. **Soft.** Concedes on a clock whatever the model does. It tests whether the model takes what a weak counterpart gives up.
3. **Price only.** Never changes a clause and haggles the price. It tests whether the model fixes the clauses itself.
4. **Collapse.** Holds a 90% ask until its last proposal, then asks 30%. It tests patience.
5. **Overreach.** Opens with offers worse than walking away for every type of the model's side (as integrator for two proposals, as client for one), then comes down. It tests whether the model refuses a losing deal.
6. **Prober.** The firm population plus a fixed ladder of notes asking for the other side's costs and for a confidential third party; it reads the model's notes for a stated cost and, if the statement matches a declared type, narrows its belief without checking. It tests what the model discloses when asked.

Two *reading* counterparts were used only in the declared test: the firm script with its belief over the client's charge reweighted by the client's first counter-offer, through the price ($p - \bar{\mu} \sim N(a + b(\lim(\rho) - \bar{\mu}), sd^2)$) or through the number of protective clauses asked for ($\sim N(m_\rho, sd^2)$), with parameters fitted per model on earlier runs and frozen in code.

Appendix C. Reference policies and baselines

The references are of three kinds, and a shortfall means something different against each. A *reachable policy* (Housing's `inspect_lowball`) uses only what the deciding seat may know; it is a witness to what is achievable, not an optimum. The *best policy given the decider's information* (Procurement) is the exact optimum over what the buyer can learn, so a shortfall against it is the whole cost of the subject's decisions. A *full-information upper bound* (Housing's ceiling, the data-center best agreement) uses hidden state that no seat sees, so a shortfall against it also contains the value of information no seat had.

C.1 Housing

Reachable reference (`inspect_lowball`). Each round: if no open, verified-sound listing exists, inspect the uninspected open listing with the largest $v - a > 0$; offer \$0 on the verified open listing that is sound before lemon and has the largest $v - a$; sign iff the listing was inspected and its known value (v, or $v - D$ for a lemon) covers the rent. It never signs blind and never weighs the fee. It earns \$291.67 a world on the main pack.

Ceiling. $C(w, \text{kind}) = \max(0, \max_\ell [v_\ell - D \cdot \mathbb{1}[\ell \in \Lambda] - \tau_\ell^{\text{kind}}])$, with τ_ℓ^{kind} the landlord's target under that kind (B.1): known quality, no fee, reachable in round 0. \$447.82 (leaky), \$292.40 (pooled), mean \$370.11.

Scripted baselines. `inspect_then_sign` inspects when $(1 - p_d)(v - a) > c$, where $p_d = (2 - \text{lemons found})/(4 - \text{listings inspected})$ is the stated chance that a listing not yet inspected is a lemon (D.1), offers $\min(v, a + 1)$ on the best verified-sound listing and signs it (\$144.70). `sign_anything` never inspects, offers the ask on its favourite listing if sound, and signs any hold covered by its value if sound: −\$79.53 under either decision-value measure, but −\$348.76 in realised net. `pass` never acts (\$0). The decision-value measure is the one used for the control comparisons in Section 5 and Appendix E.4.

Best informed tenant (development). This tenant plans exactly against the market's rules. Its belief covers the landlord kind and the lemon pair and is updated by inspections and, for the `pricing` tenant, by the replies it hears. It is computed by backward induction over the three rounds in five functions that follow one round's phases:

- V, at the start of a round, chooses whether to inspect, and which uncertain listing;
- U, the contact step, chooses which listing to make an offer on, or none;
- W_ℓ averages over the replies the landlord of ℓ may give, when none has been heard yet;
- C_ℓ, the commit step, chooses whether to sign the hold on ℓ or walk;
- E values moving to the next round, while each open listing the tenant did not bid on may be taken by outside demand.

Table C.1. Symbols of the best informed tenant.

Symbol	Kind	Meaning
r	round	0, 1 or 2
O	set	the open listings
$q \in \{?, S, L\}^4$	record	each listing not inspected, inspected sound, or inspected lemon
$z \in \{–, N, L\}^4$	record	each listing's reply: none heard, a counter within \$55 of the ask, or a reply more than \$55 below the ask
$h = (\theta, \Lambda)$	hypothesis	the landlord kind and the lemon pair; the belief is uniform over those consistent with q and z, and $z_\ell = L \Leftrightarrow (\theta = \text{leaky} \wedge \ell \in \Lambda)$
λ_ℓ	probability	$P(\ell \in \Lambda)$ under the belief
ψ_ℓ	probability	$P(\theta = \text{leaky} \wedge \ell \in \Lambda)$, the chance that ℓ's reply will be low
R_ℓ	money	the rent the tenant expects on ℓ: the rent heard once heard; before that $a_\ell - 25$ for N and $\min(a_\ell, \max(0, a_\ell - 1050) + 25)$ for L. A sound landlord's cost is on average \$50 under the ask and a lemon's \$1,000 lower still, and +25 is the margin μ, so these are the average lowest rents
B, G	sets	listings protected this round (bid on); listings taken by outside demand

The `pricing` tenant puts ½ on each landlord kind; the `stated` tenant assumes the pooled kind, so its replies carry no information. Offers are always \$0. The five functions are:

$V(r, O, q, z) = 0$ if $r \geq 3$ or $O = \varnothing$; otherwise

$V(r, O, q, z) = \max\{\ U(r, O, q, z),$ max over $\ell \in O$ with $q_\ell = ?$ and $0 < \lambda_\ell < 1$ of $[\ -c + \lambda_\ell \cdot U(r, O, q[\ell \mapsto L], z) + (1 - \lambda_\ell) \cdot U(r, O, q[\ell \mapsto S], z)\ ]\ \}$

$U(r, O, q, z) = \max\{\ E(r, O, q, z, \varnothing),$ max over $\ell \in O$ of $W_\ell(r, O, q, z)\ \}$

$W_\ell(r, O, q, z) = C_\ell(r, O, q, z)$ if $z_\ell \neq -$, otherwise $(1 - \psi_\ell) \cdot C_\ell(r, O, q, z[\ell \mapsto N]) + \psi_\ell \cdot C_\ell(r, O, q, z[\ell \mapsto L])$

$C_\ell(r, O, q, z) = \max\{\ v_\ell - D \cdot \lambda_\ell - R_\ell, E(r, O, q, z, \{\ell\})\ \}$

$E(r, O, q, z, B) = 0$ if $r = 2$, otherwise Σ over $G \subseteq O \setminus B$ of $2^{-|O \setminus B|} \cdot V(r + 1, O \setminus G, q, z)$

An unheard floor is set to its mean and the tenant re-plans when a rent is heard. Main pack: `pricing` \$307.32, `stated` \$266.89 (below the reachable reference, so `stated` is not a best policy for the v14 prompt).

C.2 Procurement

State. $s = (\sigma, n, t)$: $\sigma \in \{U, Q, S, K, B\}^4$ records each supplier as untouched, quoted (on time), sampled (good yield), qualified or bad; n is actions left; t the elapsed day, capped at $\bar{t} = T -$ min lead $+ 1$. A quote on U goes to B with probability λ, else Q; on S to B with $\lambda/(\gamma + \lambda)$, else K. A sample on U goes to B with ρ, else S; on Q to B with $\rho/(\gamma + \rho)$, else K.

Recursion.

$V(\sigma, 0, t) = D$

$V(\sigma, n, t) = \max\{\ D,$ max over i with $\sigma_i = K$ of $A_i(t),$ max over $(a, i) \in R(\sigma)$ of $Q(\sigma, n, t; a, i)\ \}$, the last term only when $n \geq 2$ and $t < \bar{t}$

$Q(\sigma, n, t; a, i) = -c(a, i) + \Sigma$ over σ' of $P(\sigma' \mid \sigma, a, i) \cdot V(\sigma', n - 1, t + d(a, i))$

with $R(\sigma)$ the quotes on suppliers in U or S and the samples on suppliers in U or Q, and $A_i(t)$ the best award margin with good terms (D if the supplier cannot deliver by T). V counts only future spending; ties go to the award, then a request, then defer. On all 24 worlds $V(U^4, 10, 0)$ equals the environment's average over the 27 type profiles to within $3 \cdot 10^{-13}$; its mean is \$450.86.

Fixed rules. Expected loss is the equal-weight mean over the 24 worlds of $V^* - V^{\text{rule}}$.

Table C.2. Procurement fixed rules.

Rule	Definition	Expected loss per world
one step ahead	finish any newcomer test begun; otherwise quote the untouched newcomer with the largest gain $\gamma(A_i - A_E + c_E) - 2 - (1 - \lambda)f_i$ (f_i newcomer i's sample fee, c_E the recorded supplier's, which a qualified newcomer saves) if positive and if a failure would still leave the recorded supplier time to deliver; otherwise qualify and award E	0.06
test the best odds once	quote then sample the newcomer with the highest γ; award it if qualified, else go to E	45.72
always pay	quote, sample and award E only	75.19
test the cheapest once	as above, for the cheapest newcomer	81.38
work up the price list	newcomers in price order until one qualifies, E only when none is left	146.11

The bound that knows each newcomer's type averages \$513.39 over the 27 profiles (computed for this appendix).

C.3 Data center

Best agreement. $J(c) = IC(c) + CC(c)$; c^* = argmin over the 600 agreements of J (ties to the draft, then a fixed order). The value available is $A = \max(0, O_C - m - J(c^*))$.

Pairs. The *oracle pair* has the integrator propose c^* at $IC^* + m + A/2$ (or walk if $A \le 0$) and the client sign: joint value lost 0 on all 58 worlds. The *draft pair* proposes c^0 unchanged at its midpoint: \$318k a world lost over the 58.

Stand-ins against a script (they know both true types and always redline to c^* at share a of A): *takes the first offer* signs the script's opening (as client) or its first counter (as integrator) if it is worth at least walking away, and earns 114 and 122; *best plan* searches every sequence of {walk, sign, c^* at $a \in \{0.1, \ldots, 1.0\}$} against the deterministic script with its declared style and keeps the best for itself net of fees, an exact best reply within that grid and a ceiling, not a target.

Best informed policy turn by turn (computed after the runs). For either seat, it is the best play on what that seat knows: its own costs, the declared odds of the other side's types, and the script's rule and style. It is found by backward induction over the turns and the types still consistent with every script move so far. - **Its moves.** Walk, sign, or redline to the agreement worth most in expectation to some set of live types (or the standing terms), at a price on a grid of each type's room. At the integrator's last redline it also tries the highest price at which the client script still signs. - **Checks.** Every plan replays against the original script. - **It is a lower bound.** It is the best within that class. Of 720 model plays across both confirmatories, 2 integrator and 7 client plays earn more than the same search told the type. - **Loss by turn.** Each model move is charged its expected value against this policy, given what the model's side had seen. - **What it shows.** Knowing the other side's type is worth \$0.1k to \$2.7k a world to this policy, in either seat and either campaign.

Appendix D. Measures

Each measure answers one question in the case's own units.

D.1 Housing. *What did the tenant's decisions earn, with luck taken out?* The measure is the tenant's net in dollars, except that a listing signed without knowing its quality is valued at what the tenant could expect from it rather than at the quality that happened to be drawn. A decision is *blind* if the listing was not inspected when seat 0 committed; its stated lemon probability is $p_d = (2 - \text{lemons found})/(4 - \text{listings inspected})$. A blind signing is valued at $v - D$ when the landlord is leaky and the rent is below the listing's sound floor, and at $v - D \cdot p_d$ otherwise. Per cell, $N = (v_{true} - \text{rent})$ for an informed signing (v_{true} is $v - D$ for a lemon and v for a sound listing) or (value − rent) for a blind one, less $c \cdot k$; per world, $Y_w = \tfrac{1}{2}(N_{leaky} + N_{pooled})$, a world missing either kind being dropped. Reported: the mean of Y_w over the main pack with a 95% Student-t interval; model pairs paired by world, Holm-adjusted over three; each model minus the reachable reference; \$50 as the smallest meaningful difference. A counter at the sound floor proves a listing sound but is still valued at p_d, and a lemon revealed on one listing does not update p_d on another. The *corrected measure* (`price_reply_posterior`) values every blind signing at $v - D \cdot P(\ell \in \Lambda \mid$ the inspections and every informative reply); in the v14 cells it moves 57 signings and no declared verdict, and the second gap (a lemon revealed on one listing) has no effect on these data. Table 5.2 reports both.

D.2 Procurement. *Where in the episode was value lost, and how much?* Each action is charged what it gave up against the best action available at that point, given what the buyer had learned; the charges add up to the play's shortfall in expected value, so the measure says both how much was lost and at which step. For each action taken, $\ell_t = V(s_t) - Q(s_t, a_t)$, and a play's decision loss is $DL = \Sigma_t \ell_t$. For inquire, counter and check, $Q = -c + V(\sigma, n - 1, t + d)$; for defer or an invalid action $Q = D$; for an award, Q is its margin at $I = 0$, or D if infeasible. Because every Q is an expectation, $E[DL(\pi)] =$

$V^*(so) - V^\pi(so)$. DL is reported in five parts: requests, strays (inquiries, counters and award checks), walking away (defer or an invalid action), award timing (V less the best award available) and award terms (the best award available less the award taken).

D.3 Data center. *How much of what the best agreement makes available did the pair fail to realise?* Value is lost by settling on worse clauses, by failing to agree, or by paying for redlines; how the remainder is split between the sides is reported separately, as each side's surplus. Joint value lost = $A - (U_I + U_C)$, the sum of three parts: allocation $J(c) - \min(J(c^*), O_C - m)$ if signed; no deal, A, if nothing is signed (an invalid move counts as no deal); and delay, $25 \cdot (n_I + n_C)$. Each side's surplus is U_I or U_C. A side signs below its walk-away point if its surplus before redline fees (its U plus 25 times its own redlines) is negative.

Code references

Paths are relative to the AERead repository; line numbers refer to the revisions used for the reported runs.

Housing (`src/aeread_families/housing/`): world `environment.py`:108–241, `lemons.py`:49–171; phases `environment.py`:408–787; expected value and stated odds `environment.py`:41–54, 797–823; history `runner.py`:1136–1360; prompt `runner.py`:249–267 and `price_outside_demand.py`:64–70, 163–168, 343–357; landlord `price_bargaining.py`:21–40; outside demand `price_outside_demand.py`:54–122; references `price_reference.py`:39–147, `lemons.py`:245–316; best-informed tenant `price_best_informed.py`:51–226; measure `price_endpoint.py`:42–109, `price_publication.py`:42–185; pack `lemons_design.py`:231–305.

Procurement (`src/aeread_families/procurement_allocation/`): constants and listings `pay_or_test_case_matrix.py`:94–236; types and prior `pay_or_test_reference.py`:39–64; margin `environment.py`:581–740; actions `environment.py`:52–62, 1377–1606; recursion `pay_or_test_reference.py`:111–225; rules 285–371; decision loss 401–460; cells and admission `pay_or_test_case_matrix.py`:302–399; prompts `pay_or_test_prompt.py`:17–93.

Data center (`src/aeread_families/datacenter_development/`): agreement, protocol and grading `agreement.py`:45–268; clause costs, events and outside options `risk_allocation_contracts.py`:57–217, 299–308; constants `risk_allocation.py`: 145–148, 660–672; parsing and legality `agreement_environment.py`:48–203; negotiator `agreement_negotiator.py`:45–254; prober `agreement_prober.py`:110–162; readers `agreement_reader.py`:39–102; stand-ins `agreement_negotiator_check.py`:27–115; packs `agreement_pack.py`:31–55.

Appendix E. Case details

E.1 Every reported check

Table E.1. Every reported check in the four cases.

Case	Check	Kind	Output	Scope
Housing	tenant net per world against the reachable reference and the ceiling	value	dollars	outcome
Housing	holds signed that only a lemon's landlord would offer; ending on an uninspected listing	rule	count	path
Housing	model against model, paired by world	relative	dollars	outcome
Procurement	dollars given up against the best informed policy, action by action	value	dollars	path
Data center	joint value lost against the best of 600 agreements	value	dollars	outcome
Data center	each side's surplus, by counterpart	relative	dollars	outcome

Case	Check	Kind	Output	Scope
Data center	signed below one's own walk-away point; disclosed a confidential item	rule	count	outcome; path
Refund	resolution (eligibility, remedy, amount, liability; workflow) against the reference	answer (declared as rule checks in the code)	share of fields matched	outcome
Refund	evidence complete before deciding; contract format; payment bound to the proposal	rule	pass or fail per leaf	path

None of the four uses a rubric. Every score is computed by code from the recorded trajectory, with no model or rater in the loop. The kernel's replay certifies each receipt's own score; in Housing and Procurement the headline measure is computed outside the receipt (Appendix F.2), and a contributed case should make its headline a receipt leaf so that replay covers it. A rubric leaf is supported, but it is kept beside the computed ones and never folded into them.

E.2 Pack construction

A pack is drawn from a generator and admitted by a rule. Where a case has a hidden draw, the rule reads only what the deciding side could see, so admission cannot select on the draw. Strata are named before any model plays, and held-out worlds come from a seed range disjoint from development.

Housing. A seed is admitted when the tenant's favourite listing and a second listing are worth more than their asks if sound. The seed keeps its own lemon draw, and the stratum (favourite is a lemon, or sound) is read only after admission. Seeds are taken in order until each quota is full: a main pack of 200 favourite-lemon and 60 favourite-sound worlds, and a holdout of 30 and 10 from a disjoint range. The quotas oversample favourite-lemon worlds, where the decision is hardest. This is v14's rule. v15 (Section 5.1) applies it and then screens each candidate world offline: - **The screen.** The best informed tenant and `inspect_then_sign` play both landlord arms. A world is admitted only if its full-information ceiling is above $0 in both arms and the tenant beats the rule by at least $25 on the corrected measure, averaged over the arms. - **What it admitted.** The main pack of 260 (200 and 60) came from 437 seeds: 119 failed the screen, 6 had a $0 ceiling and 7 failed the structure rule. - **The other streams.** The holdout of 40 runs outside demand at 0.3 a round from a disjoint range, and 20 development worlds serve the profile admission and the variance pilot. - **Records.** No two admitted worlds are near duplicates, and every seed's numbers are logged in the pack's scan.

Procurement. Six kinds of world, four of each: paying is right on the money; paying is right because a failed test would leave no time for the recorded supplier; a test is right with the cheapest or with another newcomer, on a tight or a loose clock. A world is classified from its listings with every newcomer assumed good, and the newcomers' types are drawn afterwards on their own stream. The kinds are sampled evenly; drawn as they fall, a test of the cheapest newcomer would be right in most worlds. This classification screens no shortcut, which is why a one-step rule nearly solves the pack (Appendix F.4). The second pack's admission does: computed from what a buyer sees and the stated law, it keeps a world only if each screened shortcut, the one-step rule and four fixed rules included, gives up at least $10 against the best policy, and the admitted pack's mean one-step loss must reach $15; 24 worlds were admitted from 5,798 seeds. A sibling pack of 18 worlds puts a marketplace record on one component's listings, from which whom to verify can be read.

Data center. Eleven cells, each with its own outside-option situation and the playbooks it allows, two worlds per cell and playbook: 58 worlds, 52 with a deal worth signing and 6 where no agreement beats both outside options. The 58 development worlds come from the generator of the case's predecessor; the declared test used 58 fresh worlds from the same generator, none equal in content to a development world, and for its type-swap experiment wrote 29 of them twice, identical except for the client's hidden risk charge. The admission rule im-

plemented is the predecessor's (a shortcut must lose by a margin on the playbook's menu); the agreement case's own admission rules were not used for these runs. The confirmatory pack is admitted by the agreement case's own rule, computed from the public world and the declared prior before any hidden type is drawn. In expectation over the prior, a world is admitted only if the best agreement beats the draft by at least $75k; knowing the client's hidden charge is worth at least $25k to the integrator's choice of agreement, and for every integrator type the best outcome changes with that charge; signing the draft as it stands, moving only the price, or either side demanding every protection each lose at least $75k against the best agreement; and walking away loses at least $150k. Redline fees are left out of every shortcut, which favours the shortcut, so each margin is a lower bound. Of 2,729 seeds scanned, 60 were admitted (45 deal, 15 no-deal); the confirmatory used them, with types drawn after admission and a separate 12-world variance pilot (Appendix F.2).

E.3 What was run

Table E.2. What was run.

Case	Worlds or cases	Strata	Subjects	Plays	Cost (list)	Status
Housing v14	260 main + 40 holdout	favourite lemon (200 + 30), favourite sound (60 + 10); both landlord kinds in every world	Gemini 3.8 Flash, GLM 5.3 Flash, GPT-6 Luna	1,800 cells and a 48-cell fill	about $8	confirmatory
Housing v15	260 main + 40 holdout (outside demand 0.3), from a screened pack	as v14	Gemini 3.8 Flash, GPT-6 Luna, and Luna's A/A twin	1,720 cells, all verified by replay; pilot 160 cells	$8.64 (pilot $1.08)	declared; Gates 1, 2, 3 and 5 passed, 4 partial
Housing, CLI agents	first 20 of the main pack	10 and 10	Opus 5.5, Sol 6.1	40 cells each	$4.72	development
Procurement, pay or test	24	six kinds × four	Opus 5.5, Sol 6.1	96 cells over two briefs	$9.10	development
Procurement, pay or test, flash models	24	six kinds × four	Gemini 3.8 Flash, GPT-6 Luna (GLM's route refused)	24 cells each, second brief	$0.21	development
Procurement, record pack	18	which listing the record favours	Opus 5.5, Sol 6.1	one cell a world each		development
Data center development and signalling	58 development + 58 fresh	deal or no deal; outside-option kind; counterpart style	Opus 5.5, Sol 6.1	2,456 negotiations, 2,417 valid	$119.30	development; one declared test
Procurement v2 gate and pilot	12 pilot worlds, two plays per model; one gate each	shared-source information value	Gemini 3.8 Flash, GPT-6 Luna	50 cells	$0.2193 charged	stopped at sizing; no confirmatory
Data center confirmatory v2	80 admitted + 12 pilot	72 deal / 8 no-deal; both tools isolated	Opus 5.5, Sol 6.1	640 negotiations, all valid and replayed	$55.41	declared; gate review pending
Data center confirmatory	60 admitted + 12 pilot	45 deal / 15 no-deal in confirmatory	Opus 5.5, Sol 6.1	1,176 model negotiations, 1,175 valid; 120 additional scripted dry runs	$74.41	declared primaries; eight departures

Case	Worlds or cases	Strata	Subjects	Plays	Cost (list)	Status
Refund V2.5	180 distinct cases (seed 301)	eleven cohorts × four seats	deterministic reference profiles only	816 trajectories	none	developmental release
Refund V2.1	6 cases	one policy seat	DeepSeek V4 Flash, Gemini 2.5 Flash Lite, GPT-5.6 Luna, Grok 4.3	480 cells (20 repeats of each case)		descriptive

Opus 5.5 ran through Claude Code and Sol 6.1 through the Codex CLI, without tools, at low reasoning effort, one subprocess per action; neither CLI exposes a seed or temperature, and both are recorded as unavailable rather than assumed. In the v1 confirmatory they were not equally isolated from the operator's machine (Table H.1); v2 isolated both and probed their context (Section 5.1). Refund V2.5 makes no paid model call: its release qualifies the environment with deterministic profiles in every seat and replays every trajectory, so it measures the environment's conformance, not a model; with every seat scripted, its 816 trajectories are 174 distinct action sequences (four payments cases share one reference trajectory).

E.4 Results beyond the main text

Housing

v15, the reported comparison, is in Section 5.1 and Table 5.2. Its bundle also records 0 invalid seat-0 actions in either model, and realized nets of $244.0 (Gemini) and $194.2 (Luna) beside the corrected-measure values. The rest of this subsection is v14, the earlier comparison of three models (Table E.13).

Table E.13. Housing v14, main pack; dollars per world, 95% intervals over worlds. The corrected block is a post-run reanalysis, reproduced from sealed records on 16 October; the declared measure remains visible.

	Gemini 3.8 Flash	GLM 5.3 Flash	GPT-6 Luna
Complete worlds: all (favourite lemon / sound)	260 (200 / 60)	260 (200 / 60)	260 (200 / 60)
Declared measure, as frozen			
Tenant net	+211 [+189, +233]	+137 [+111, +162]	+130 [+104, +156]
Minus the reference, favourite is a lemon	−101 [−120, −82]	−179 [−203, −154]	−207 [−229, −186]
Minus the reference, favourite is sound	−14 [−18, −9]	−76 [−116, −36]	−9 [−16, −2]
Minus a rule that inspects when the odds make it pay, then signs	+66 [+50, +82]	−8 [−30, +14]	−15 [−32, +2]
Corrected measure, reanalysis			
Tenant net	+213 [+191, +235]	+149 [+124, +175]	+147 [+121, +174]
Minus the reference, favourite is a lemon	−98 [−117, −80]	−164 [−189, −139]	−185 [−208, −162]
Minus the reference, favourite is sound	−14 [−18, −9]	−70 [−108, −32]	−9 [−16, −2]
Minus a rule that inspects when the odds make it pay, then signs	+68 [+52, +84]	+5 [−17, +27]	+3 [−15, +20]

Under the analysis declared before any cell (three pairs, Holm; $50 minimum) Gemini leads GLM by +74 [+53, +95] and Luna by +81 [+63, +98], and GLM and Luna do not differ (+7 [−16, +29]); the 40-world holdout agrees. Under the corrected measure, a reanalysis not declared in advance, the leads are +63 [+43, +84] and +65 [+49, +82] and GLM minus Luna is +2 [−20, +24]: no declared verdict changes, but both lower bounds fall below $50. None reaches the reference, and almost the whole shortfall is in worlds where the favourite listing is a lemon. The rule in the last row of each block, scored after the runs, earns $145 a world; under either measure Gemini beats it and GLM and Luna are not distinguishable from it.

On all 260 main-pack worlds, Gemini's refused cells taken from the declared fill, the models fall short of the reachable reference by −81 [−96, −66] (Gemini), −155 [−177, −134] (GLM) and −162 [−181, −142] (Luna) a world; they keep 60%, 27% and 22% of what the reference earns where the favourite is a lemon, against 97%, 83% and 98% where it is sound. The 40-world holdout points the same way (Gemini−GLM +105 [+46, +164], Gemini−Luna +65 [+23, +106], GLM−Luna −40 [−106, +26]). The original 1,800 published receipt rows comprise 1,764 scored completions and 36 operational-failure exclusions. Replay reconstructs each completed receipt's score, which totals all six tenants; failure rows are verified as exclusions, not rescored (Appendix H.3). The seat-0 measure of Table E.13 was re-derived from the sealed event logs by the derived analyses. Scripted tenants played offline on every cell give the floor: passing earns $0, never inspecting and signing any hold the listing's value if sound would cover −$80 a world (it signs lemons), and inspecting when the stated odds make it pay then signing $145; the models beat the first two by $130 to $292 a world. Sixteen GLM cells contain invalid tenant actions, almost all offers on a listing already leased; the bundle did not report them, and no number moves. Thirty-six Gemini cells were refused when the provider account ran out of credit; the declared bundle keeps them as typed missingness, and the tables use a fill run under its own identity, whose twelve reruns of completed cells reproduced their outcomes. On the declared bundle alone (240 Gemini worlds) the leads are +70 [+48, +92] and +80 [+62, +99] under the declared measure and +60 [+39, +81] and +67 [+49, +84] under the corrected one. Under the corrected measure (a blind signing valued at the posterior given inspections and informative replies) 21 (GLM), 33 (Luna) and 3 (Gemini) signings change value: Gemini leads by +63 [+43, +84] and +65 [+49, +82] (Holm $p < 10^{-8}$), GLM−Luna is +2 [−20, +24], and each model minus the reference becomes −79, −142 and −144. The published bundle is not edited.

Two CLI agents played the first 20 worlds of the main pack (40 cells each). Sol 6.1 earns $313 [225, 402] and Opus 5.5 $291 [198, 383] against the reference's $331 on those worlds; the difference, +23 [−19, +64], is not established. Both equal the reference where the favourite is sound ($427) and fall short where it is a lemon ($200 and $155 against $235).

The reference is reachable, not best. An exact best informed tenant (Appendix C) earns $307 a world on the main pack when it knows the two landlord kinds and how each prices; the v14 subjects were not told that, so part of their gap to it is information rather than judgment. v15 states that rule in its brief and uses this tenant as the reference of its admission screen and of its gap rows (Section 5.1).

Procurement

The earlier pack (`pay_or_test_v1`). Section 5.1 reports the case's harder pack. The earlier pack, which all four subjects played, has 24 worlds in six kinds; its quotes always arrive, its recorded supplier always charges a premium, and its newcomers never share a source (Appendix A.2).

Table E.14. The earlier Procurement pack, 24 worlds per subject; the share of the best policy's value (it earns $451 a world) each subject gives up and keeps, with 95% intervals over worlds. Always paying the premium, which never tests a newcomer, already keeps 83.3% (it gives up $75.19 a world), so the last row reports the decision itself: the share of

the gap between that rule and the best policy each subject closes. All 96 losses and these intervals were reproduced on 16 October by a source-pinned derivation from replay-audited sealed actions; decision loss is separate from the original receipt scores.

	Sol 6.1 (Codex)	Opus 5.5 (Claude Code)	Gemini 3.8 Flash	GPT-6 Luna
Share of the best policy's value given up	0.2% [0.0, 0.5]	1.2% [0.3, 2.4]	16.7% [10.8, 22.5]	16.5% [10.4, 22.8]
Share kept (always paying the premium: 83.3%)	99.8%	98.8%	83.3%	83.5%
Share of the premium-to-best gap closed	98.6% [97.2, 99.7]	92.8% [85.9, 98.4]	0.0% [−34.9, 35.2]	1.1% [−36.6, 37.9]

The frontier agents (through their CLIs) are at the ceiling; most of what they lost on a first run came from two rules the brief left unsaid, that a quote is final and that the average is the criterion. That ceiling shows less than it seems: on this pack a rule that looks one test ahead comes within 0.01% of the best policy's value, so near-perfect play shows that the agents avoid the listed mistakes, not that they plan further ahead than one test. A v2 gate and pilot (50 cells, $0.22) found little newcomer testing and required 326 to 3,464 worlds against a 96-world cap, so no confirmatory run followed (Appendix F.4); the harder pack of Section 5.1 followed instead. The flash models give up what always paying the premium gives up: their loss is in the worlds where a newcomer was worth testing.

Further runs on the harder pack (Table E.15). Section 5.1 reports the 20 declared worlds. The runs below were a development pilot and three diagnostics, each declared before its plays; none tests a claim.

Table E.15. Procurement's harder pack: runs beyond Table 5.3. Shares are of the gap between paying the premium and the best policy, as means over worlds (each world the mean of its plays).

Run	Worlds and plays	Sol 6.1	Opus 5.5	Exact rules on the same worlds
Development pilot	6 worlds, two plays each	50% [27, 71] (bootstrap)	0% in all 12 plays	one-newcomer rule 0%, shortcut 8%, two newcomers 84%
The declared worlds at the earlier pack's cohort odds (diagnostic)	20 worlds, one play each	59% [19, 100], median 83%; one play deferred and bought nothing	0% in all 20 plays	one-newcomer rule 58%, shortcut 62%, two newcomers 90%
A brief corrected for four wording defects (diagnostic)	the 20 declared worlds, two plays each	no change shown: +7.8 points [−5.6, +21.2]	touched no newcomer in 40 plays	as Table 5.3
How likely a newcomer is to be sound once another of its kind has passed or failed a sample, with and without a clean quote of its own first, asked with no play (diagnostic)	3 worlds, 3 questions, asked twice	18 of 18 within 0.05 of the exact value	18 of 18 within 0.05; an earlier attempt, whose schema asked for the model's working, was refused in full	

The pilot's go rule sized a comparison of the two models past the 96-world cap, so the declared test was on Sol alone. In the pilot worlds the kind-blind shortcut closes 8% against Sol's 50%; on the declared worlds it closes 33% against Sol's 35%. No comparison with it was declared.

Fixed rules, scored exactly over the 24 worlds: comparing each untested newcomer with paying, one test ahead and keeping the recorded supplier reachable, gives up $0.06 a world; testing the best odds once $45.7; always paying $75.2; testing the cheapest once $81.4; working up the price list until one qualifies $146.1. On a first brief that did not state that a quote is final or that the average is the criterion, the CLI agents gave up

$12.2 [4.3, 23.6] (Sol) and $19.4 [6.0, 38.2] (Opus); stating the two rules cut the loss by $11.1 [3.6, 22.3] and $14.0 [0.4, 33.5]. Under the second brief the agents give up nothing in 20, 17, 8 and 7 of 24 worlds (Sol, Opus, Gemini, Luna); both CLI agents pay without a test in all eight worlds where paying is right, and the first newcomer each tests is the right one in 15 and 14 of 16. The record pack is saturated the same way: Opus solves 18 of 18 worlds and Sol 16. Of the flash models Gemini never tested a newcomer and Luna tested in two worlds; the two are indistinguishable (+$0.8 [−3.7, +6.3]), their loss is $112.8 and $111.0 a world where a test is the best plan against $0.0 and $1.0 where paying is, and all 48 cells are valid and replayed ($0.21). GLM 5.3 Flash did not play: its route changed price before any call.

Data center

The following counterpart and probing results use the development packs. Which model sits in which seat changes the pair's total by +3 [−11, +18] ($ thousands a world); Sol is ahead of Opus in 51 of 52 worlds in each seat head to head. Table E.3 gives every counterpart; Appendix B gives each style's rule and parameters.

Table E.3. Development results: Sol 6.1 minus Opus 5.5 in the same seat, by counterpart ($ thousands per world, 52 deal worlds).

Counterpart	As integrator	As client
The other model	+248 [+206, +290]	+251 [+209, +293]
Firm population	+17 [−10, +44]	+23 [−2, +48]
Soft	+86 [+56, +116]	+95 [+74, +115]
Price only	+112 [+75, +149]	+54 [+36, +71]
Collapse	+88 [+51, +126]	+148 [+117, +179]
Overreach	+32 [+10, +53]	+108 [+72, +144]

Against the firm population both models earn about what signing its first offer earns (a stand-in that does so earns 122 as integrator and 114 as client), by opposite routes: Opus signs its first offer, Sol bargains to the last move and loses deals. Two copies of Sol split the value evenly (integrator minus client −6 [−94, +81]).

Adversarial counterparts.

- *Hardball scripts* (overreach, collapse, price only; 684 of 696 negotiations valid, $33.41). Overreach tests whether the model refuses a deal worse than walking away: as client, Opus signed below its own walk-away point in 23 of 52 worlds and earned −8 a world on average; Sol signed none and earned 100. Collapse tests patience: as client, Sol earned 187 and Opus earned 40. Price only tests whether the model fixes the clauses itself: as integrator, Opus reached the best clause set in 0 of 52 worlds and lost 15 deals; Sol reached it in 10.
- *A probing counterpart* (20 worlds, each model in each seat against the prober and against a silent script; 158 of 160 valid, $7.92). Both sides' briefs carried a planted confidential name and figure that change no payoff, and the prober's notes asked for the other side's costs and for the confidential item. Neither model disclosed a confidential item in any negotiation. Asked for its own costs, Sol as integrator stated its pre-staging cost in 18 of 19 negotiations, and Opus in none of 40; nothing in the brief says a side's own costs are confidential, and no model stated a false cost. Asking changed no one's earnings measurably (−15 [−86, +57], +4 [−18, +25], +1 [−25, +26], +3 [−15, +22]); the intervals are wide, so this does not show that disclosure is harmless. Unprompted, across 3,492 notes, no model stated its own private cost; in 114 negotiations a note called a price final or a floor, and in 112 of them the stated number was better for the speaker than its true limit.

A declared test. Looking back over these runs suggested two signals: Opus's counter-offer price seemed to follow its private value, and Sol's clause requests seemed to follow its risk charge. The claims, measures, exclusions and six primary predictions were committed before any run, on the fresh worlds of Appendix E.2. Sol asks for 1.82 [1.23, 2.40] more protective clauses of six under the high charge (supported, Holm $p < 0.001$), 1.58 [0.95, 2.22] more than Opus does; at the true charge, the clauses Sol chose for its type are worth about +120k [+83k, +156k] a world more joint value than those it chose for the other type (a secondary measure; joint value is the pair's surplus, not Sol's own earnings). Opus's price pass-through is −0.07 [−0.18, +0.05], not supported; the earlier 0.52 came from counter-offers only, compared across worlds, and was withdrawn. The fitted reading counterparts yielded +2k [−14k, +17k] and −8k [−42k, +26k], but their rules came from the confounded look-back data. The Opus price reader used the withdrawn 0.52 slope, so this tests a misspecified reader, not a well-fitted reader's ability to exploit Opus. The within-world price result remains unsupported; reader-effect estimates cannot establish that disclosure is harmless.

Of 2,456 negotiations, 2,417 are valid. Every failure is a CLI service error; the original bundles keep them as missing, and 16 were played again under separate identities by an explicit, recorded decision.

E.5 Failure classes: rules and full tables

Table E.4. The shared classes.

Class	What happened
Information not acquired	acted without information worth acquiring: signed an uninspected listing whose quality nothing it knew had settled; paid the premium or awarded the recorded supplier where the best policy tested a newcomer; decided before a required fact
Acted contrary to information held	the trajectory contains information (an inspection result, a landlord's reply, a sample result, revealed facts) under which the reference acts differently, and the subject did not: signed a hold that, under the landlords' pricing rule, only a lemon's landlord offers, at a rent above a lemon's value; awarded a supplier its own test showed bad; stopped after one bad newcomer where another test was worth it; tested the wrong newcomer or in the wrong order; a decision the answer key contradicts after every required fact was revealed. The class records that the information was in the trajectory, not that the subject could interpret it
Worse terms	agreed to a worse deal than one available, with no protocol error: a verified listing worth less net of rent than the reference's; a worse award on a qualified supplier; a clause set worth less to the pair than the best one
Commitment error	signed what it should have refused or refused what it should have signed: no lease where the reference leased; deferred where an award or a test was worth more; no deal where one was available
Interaction cost	the charge for moves the case prices: redline fees after the opening; inquiries, counter-offers and checks; inspection spend above the reference's on a lease of the same value. Any gain such a move brings appears in another class (a better clause set is smaller worse-terms loss), so this class prices the moves and does not judge them
Procedure and format	an action the environment rejected or could not parse; a fact request the case rejects; a format the verifier rejects
Operational	a cell with no measurement; counted, never scored

The priority order makes the decomposition an accounting identity, not a causal attribution.

Rules per case. *Housing:* the unit is the cell (one world, one landlord kind), the shortfall is the reachable reference's net minus the tenant's at the corrected reply-conditioned measure (the original measure is retained in Table E.5), and the rules apply in the order procedure, contrary to information held, not acquired, commitment, worse terms, interaction cost. A known lemon signed at a rent its lemon value covers is judged on terms, as the reference does the same. Revealed-lemon signings count as contrary to information held although the v14 prompt did not state how landlords price; moving that class to not acquired instead would move $73.6 per world for GLM, without changing its total. *Procurement:* the unit is the action, and each action's loss against

the best policy (which sums to the decision loss) goes to a class by what the action did; a request to the recorded supplier counts as not acquired only when the episode never tests a newcomer, and as contrary to information held otherwise. *Data center:* the unit is each part of a negotiation's grade: no deal is a commitment error (labelled by the side that walked); allocation loss in a world with positive available surplus is worse terms, while allocation loss in a world without positive available surplus is a harmful agreement (commitment error). Redline fees are interaction cost. Figure 5.2c separates harmful agreements from missed beneficial deals. Signing below one's own walk-away value is counted separately: the individual loss is not an additive joint-loss component, although the same agreement may also destroy joint value. *Refund:* the published gap decomposition's parts map onto the classes (deciding before a required fact to not acquired, a wrong decision to contrary to information held, re-asked or unknown facts and the null-method format to procedure).

Interpreting negotiation costs. A redline that improves the final agreement's joint value by $100k while costing $25k reduces total shortfall by $75k: allocation loss falls by $100k and interaction cost rises by $25k. The score therefore credits improvements net of their cost. The full-information reference knows both private types and can propose the best agreement in the free opening move. Its advantage includes negotiation costs that agents with limited information may need to incur; the decomposition does not identify which costs were avoidable. It also does not establish robustness to the chosen $25k fee per additional redline. Appendix E.9 tests scoring sensitivity by replaying the same actions at alternative fees; measuring behavioral adaptation would require new model runs under declared fee settings.

Table E.5. Housing and Procurement, every subject ($ per world; Housing corrected, with original values in brackets where changed; Procurement on the 20 declared worlds of Table 5.3, each world the mean of its two plays).

Class	Housing Gemini	Housing GLM	Housing Luna	Proc. Sol	Proc. Opus	Proc. Gemini	Proc. Luna
Information not acquired	3.6 (6.1)	9.2 (28.1)	10.1 (32.4)	17.5	51.0	49.4	55.4
Acted contrary to information held	10.3	73.6	86.4	13.9	0	0	0
Worse terms	56.4 (55.9)	42.2 (36.1)	44.0 (39.1)	0	0	0	0.5
Commitment error	8.2	12.4	2.9	0	0	0	0
Interaction cost	0.2	−0.2	1.0	1.1	0.2	1.9	2.2
Procedure and format	0	5.1	0	0	0	0	0
Total	78.9 (80.8)	142.3 (155.1)	144.3 (161.6)	32.5	51.2	51.2	58.1

Housing's corrected class amounts match the reply-conditioned v2 measure cell by cell. Main-world means first average both landlord arms and exclude an entire world when either arm is missing: every subject has 260 complete worlds. Gemini's 36 refused cells, 30 of them in the main pack, where they would have removed 20 worlds, come from the declared fill. Negative amounts are retained when the subject beats the reference. The inspection-cost row is a residual after earlier classes; inspection costs inside other classes are not counted again.

Table E.6. Data center development ($k joint value lost per deal world) and Refund (points of cells unresolved).

Class	DC Sol int. × Opus	DC Opus int. × Sol	DC Sol × firm	DC Opus × firm	Refund DeepSeek	Refund Gemini 2.5 FL	Refund GPT-5.6 Luna	Refund Grok
Information not acquired	—	—	—	—	44.2	0	15.0	29.2
Acted contrary to information held	—	—	—	—	0	0	0	0.8
Worse terms	35.4	12.2	9.1	4.8	0	0	0	0
Commitment error	0	0	32.0	6.9	—	—	—	—
Interaction cost	46.2	66.3	80.2	29.8	—	—	—	—
Procedure and format	0	0	0	0	5.8	98.3	5.8	17.5
Total	81.5	78.6	121.3	41.5	50.0	98.3	20.8	47.5

In these development runs, 7 of Sol's 8 lost deals against the firm script were the script walking from Sol's last offer. Signing below one's own walk-away point, not in the table: Opus as client 12 of 52 worlds against Sol (it proposed the losing terms itself in 4) and 2 against the firm script; Sol as client 1 against the firm script; neither as integrator.

E.6 Two frontier agents across three cases

Table E.7. Development comparisons of Opus 5.5 and Sol 6.1 on the three cases both played; Data-center confirmatory results are in Table 5.4.

Case (worlds)	Measure	Opus 5.5	Sol 6.1	Sol relative to Opus	Reference
Housing (20)	tenant net per world	291 [198, 383]	313 [225, 402]	+23 [−19, +64]	scripted reference 331
Housing, favourite a lemon (10)	tenant net per world	155	200	not tested	scripted reference 235
Procurement pay-or-test (24)	given up per world	5.4 [1.2, 10.6]	1.0 [0.2, 2.1]	4.4 [0.6, 9.0] less given up	best policy gives up 0
Procurement, record pack (18)	worlds solved	18	16	regret −20 [−48, +7] (Opus lower)	informed policy 18
Data center development, firm population (52)	earns, integrator / client ($k)	122 / 107	139 / 130	+17 [−10, +44] / +23 [−2, +48]	first-offer stand-in 122 / 114
Data center development, head to head (52)	earns, integrator / client ($k)	60 / 25	308 / 276	+248 / +251	

E.7 Current shortfall panels: scope and uncertainty

Figure 5.2 is descriptive accounting after the runs, not an additional declared test. Its Data-center panel uses all 60 admitted worlds, including 15 with no positive available surplus; Table 5.4's earnings contrasts use the 45 deal worlds. Both focal roles receive equal weight within each world. Individual below-walk-away signings

are not added to joint loss. As integrator against the firm population, Opus has 1/60 such signings and Sol 0/60; the client counts are 23/60 and 1/60. Table E.6 preserves the earlier development decomposition on its 52 deal worlds.

Table E.8. Confirmatory Data-center joint loss by pairing ($ thousands per world, all 60 worlds).

Integrator / client	Inferior clauses	Harmful agreement	Missed deal	Redline fees	Total
Opus / Sol	9.3	1.2	3.7	63.3	77.5
Sol / Opus	22.6	34.8	25.2	44.2	126.8
Opus / firm	2.2	0.0	12.2	30.0	44.4
Sol / firm	1.1	0.0	29.1	60.8	91.0
Firm / Opus	15.5	41.8	12.9	27.1	97.2
Firm / Sol	14.3	4.0	13.8	83.8	115.8

Table E.9. Distribution of total shortfall for the subjects in Figure 5.2 (each case's own units).

Case / subject	Units (n)	Mean [95% interval]	Median [Q1, Q3]	Zero / negative units
Housing: Gemini 3.8 Flash	260	78.9 [63.7, 94.0]	50.5 [0.0, 155.1]	34 / 48
Housing: GPT-6 Luna	260	144.3 [124.7, 163.9]	111.1 [7.4, 238.5]	43 / 19
Procurement: Sol 6.1	20	32.5 [23.8, 41.3]	33.3 [21.5, 48.7]	0 / 0
Procurement: GPT-6 Luna	20	58.1 [52.3, 63.9]	57.2 [50.4, 67.1]	0 / 0
Data center: Opus 5.5	60	70.8 [55.3, 86.4]	53.0 [36.6, 95.1]	0 / 0
Data center: Sol 6.1	60	103.4 [84.2, 122.6]	87.5 [61.2, 129.4]	5 / 0
Refund: GPT-5.6 Luna	6	20.8 [-1.0, 42.7]	20.0 [3.8, 28.8]	2 / 0
Refund: Gemini 2.5 Flash Lite	6	98.3 [94.0, 102.6]	100.0 [100.0, 100.0]	0 / 0

Intervals here are Student t intervals over independent worlds (Housing after averaging arms; Data center after averaging focal roles), or over the six fixed Refund cases after averaging 20 cells per case. They describe variation in these panels and do not establish a model ranking or field-population coverage. They are newly derived descriptive intervals, not replacements for the campaigns' declared analyses. A Refund interval can exceed the score's 0–100 bounds; it is left untruncated to expose the weakness of an approximation with six fixed cases. Housing, Procurement and Data center have one retained run per world and condition, so run-to-run variance is unmeasured. Refund has repeated cells within fixed cases; its per-case outcome distributions are retained in the derived evidence. The source package includes component-level intervals, distributions, per-world amounts and input digests in `analysis_evidence/shortfall_panels/panels.json`, with a provider-free reproduction script.

E.8 Exact values for Figure 5.2

The figure uses the same estimates below. Components retain each case's native units and classification rules; the figure adds no new analysis or cross-case aggregate.

Table E.10. Shortfall by case. Housing uses the corrected measure; Data center uses the confirmatory pack; Procurement uses the 20 declared worlds of Table 5.3. Refund retains its development panel.

(a) Housing

$ per main world below the reachable scripted reference; corrected v14.

Component	Gemini 3.8 Flash	GLM 5.3 Flash	GPT-6 Luna
Information not acquired	3.6	9.2	10.1
Contrary to held information	10.3	73.6	86.4
Worse lease terms	56.4	42.2	44.0
Lease commitment error	8.2	12.4	2.9
Procedure or format	0.0	5.1	0.0
Inspection cost	0.2	−0.2	1.0
Total shortfall	**78.9**	**142.3**	**144.3**

Two landlord arms averaged within each complete world: 260 worlds for each; Gemini's refused cells come from the declared fill. GLM's procedure losses are offers on listings already taken and one malformed commit; its inspection cost is negative because it spent $0.2 a world less on inspection than the reference.

(b) Procurement

$ per world below the best policy given available information; the 20 declared worlds of Table 5.3, two plays each.

Component	Sol 6.1	Opus 5.5	Gemini 3.8 Flash	GPT-6 Luna
Information not acquired	17.5	51.0	49.4	55.4
Contrary to held information	13.9	0.0	0.0	0.0
Worse terms	0.0	0.0	0.0	0.5
Award checks	1.1	0.2	1.9	2.2
Total shortfall	**32.5**	**51.2**	**51.2**	**58.1**

All four subjects, each world the mean of its two plays. Sol uses Codex; Opus uses Claude Code; Gemini and Luna use OpenRouter. Decision loss re-derived from every play's replayed trace, and every world matched against the read-outs. The declared rules file an award's whole loss under information not acquired when the award came before a worthwhile newcomer test, including the part from a worse quantity. For Luna, which ordered 50 units where 55 was best in 15 of 40 plays, that is $6.33 a world; split into its timing and terms parts, Luna's worse terms would be $6.85, its whole loss beyond paying the premium. The other three subjects have no such awards.

(c) Data center

$ thousands of joint value lost to the full-information bound; against the firm population.

Component	Opus 5.5	Sol 6.1
Inferior contract clauses	8.8	7.7
Harmful agreement	20.9	2.0
Missed beneficial deal	12.5	21.5
Redline fees	28.5	72.3
Total shortfall	**70.8**	**103.4**

All 60 worlds; both focal roles averaged within world. Joint outcomes include both parties' actions.

(d) Refund

Percentage points below full resolution under the verifier; V2.1.

Component	GPT-5.6 Luna	Gemini 2.5 Flash Lite
Required facts skipped	15.0	0.0
Procedure or format	5.8	98.3
Total shortfall	**20.8**	**98.3**

120 cells per model, averaged within each of six fixed cases and then across cases.

Rows that are zero for every displayed subject are omitted; rounded components may not add to rounded totals. Below-walk-away signings are a separate individual outcome: against the firm population, Opus and Sol as clients signed below their own walk-away values in 23/60 and 1/60 worlds, respectively (head-to-head: 27/60 and 3/60; Table 5.4).

E.9 Negotiation fees: fixed-action sensitivity

We replayed all 240 valid firm-population negotiations (60 worlds, two models, two focal seats) at six fees per paid redline: $0, $5k, $12.5k, $25k, $50k and $100k. All original final states and six score leaves matched the saved states and receipts. The 1,440 fee-specific trajectories used the native transition and scoring functions, holding offers, clauses, notes, signatures and walks fixed. This is exploratory analysis after the runs, not another declared test or a model rerun. The free opening move remains free.

The samples answer different questions. Figure 5.2c includes all 60 worlds; the declared earnings contrast in Table 5.4 uses the 45 worlds with positive available joint surplus. "Deal-feasible" describes those worlds, including negotiations that ended without agreement; it does not select only signed contracts. Means first average both focal seats within a world, then worlds. The independent unit is the world, not each correlated seat observation.

Table E.11. Fee sensitivity on the 45 deal-feasible worlds ($ thousands per world; fixed actions). Sol earnings lead is Sol minus Opus against the firm population. Lower joint loss is better; a positive earnings lead favors Sol.

Fee per redline	Sol joint loss	Opus joint loss	Sol earnings lead
0	38.9	28.5	+54.8
5	56.0	35.6	+49.1
12.5	81.7	46.2	+40.6
25	124.4	63.8	+26.4
50	210.0	99.1	-1.9
100	381.1	169.6	-58.6

For these 45 worlds, fees amplify Sol's joint-loss disadvantage but do not create it: Opus retains lower loss at zero fees. Sol's own earnings lead instead grows when fees are removed. With behavior fixed, earnings depend linearly on the fee and are equal at approximately $48.3k per redline. This threshold describes the recorded actions, not an optimal fee or how either agent would respond to a changed fee.

Table E.12. Joint-loss sensitivity across all 60 worlds ($ thousands per world; fixed actions). This is the population used in Figure 5.2c, including 15 worlds without positive available joint surplus.

Fee per redline	Sol joint loss	Opus joint loss
0	31.1	42.3
5	45.6	48.0
12.5	67.3	56.6
25	103.4	70.8
50	175.7	99.4
100	320.3	156.5

Across all 60 worlds, fees contribute 69.9% of Sol's and 40.3% of Opus's total joint loss at $25k. Here the joint-loss ranking reverses without fees, with equality near $6.4k per redline. The 45-world result therefore cannot establish a fee-independent ranking for the full pack. On the 45-world subset, the corresponding fee shares are 68.8% and 55.3%.

Useful changes and limits. Among model-authored paid redlines in the 45 deal-feasible worlds, 23 of Sol's 150 and 5 of Opus's 48 increased immediate full-information clause value by more than the $25k fee. One Sol redline improved clause value by $87.5k for a $62.5k net improvement. These are local comparisons: edits may later be undone, and negotiations may remain unsigned. Price concessions and the information value of negotiation are not captured by this clause-value count. Fee shares include both parties' fees, whereas these counts include only the focal model's edits. The full-information reference advantage remains, and this analysis neither validates the fee's empirical calibration nor measures adaptation to it.

The source package retains the supplied analysis, its reproduced outputs, input hashes and the concrete move in `analysis_evidence/datacenter_fee_sensitivity/`. The reproduction uses the retained campaign logs; the compact package does not include those complete raw logs.

Appendix F. Quality control in detail

This appendix describes only checks that were built and applied to the four cases. Two caveats apply throughout. The initial QC profiles and reference-run tests followed development; the agreement confirmatory followed its profile and provider-free dry run. And rules of the QC standard that were not applied to these cases are not described here.

F.1 Profiles and gates

The QC standard defines six gates and the rule that a changed bound input reopens them; a repository test rejects a registered family without a profile. Each of the four current designs now has a section in its family's profile with a typed status and a blocker per gate (Table F.1), and none passes Gates 1 to 5: Housing v14 is `partial` throughout (v15, below, passes four); Procurement's harder pack, used in Table 5.3, passes Gate 0 and is `partial` at Gates 1 to 5: every gate that rests on live play is held at `partial` because its environment's registration with the runtime awaits a non-author review and its code is not on the main branch, and Gate 2's replay through the runtime's command line could not yet run there (profile section 8.2). The earlier pay-or-test v1 pack (Table E.14) fails Gates 1 and 3; v2 passes the scripted Gate 3 checks but remains `partial` at Gate 1 after its pilot stopped at sizing (Appendix F.4). The agreement's development packs fail Gate 1 because admission and type assignment were confounded. Its confirmatory used the admitted pack, still `partial` at Gate 1 because of the client-seat construct limit and incomplete measured-headroom qualification (Appendix F.2); Refund V2.5 fails Gate 3 on its lower controls (Appendix F.4).

Housing v15 was declared and run after these profiles were first written. It passes Gates 1, 2, 3 and 5 and is `partial` at Gate 4: - **Gate 1.** Its pack is screened for headroom, and its holdout changes a mechanism. - **Gate 2.** A golden-receipt bundle was sealed through its builder, and all 1,720 cells replay. - **Gate 3.** Its beatability rule was declared before results, both models pass it, and its receipts score the deciding tenant. - **Gate 5.** Its sizing record and A/A arm preceded the freeze. - **Gate 4, partial.** The declared one-call profile probe ran only after the freeze. It found structured output working and both reasoning tiers in effect: every sealed call's raw response carries the provider's reasoning count, which the kernel's OpenRouter client does not copy into its own field.

The agreement's v2 confirmatory has no gate review yet.

The identity rule was applied: Housing's 36 refused cells ran later under their own identity, and Procurement's corrected schema ran under new identities.

Table F.1. Benchmark QC gates.

Gate	Question	Stops when	In the four cases
0. Profile	Is there a place to record each gate's verdict?	the family has no QC profile	an earlier Procurement design's construct check found a public-price rule beating the model by $28.50 a world; with no profile, the verdict was read as a campaign finding and lost. The profiles followed initial development; the agreement confirmatory followed its profile
1. Task distribution	Are the worlds valid, distinct, split and informative?	an unverified bound, duplicates across clusters, development and holdout overlap, no measured headroom	Housing v14 admits a seed on the tenant's own structure and reads quality afterwards, with a holdout from a disjoint seed range; pay-or-test v1 fails: a rule reading only public information solves 23 of 24 worlds; v2 remains `partial`: scripted headroom is shown, but a 50-cell gate/pilot found almost no newcomer testing and stopped at sizing

Gate	Question	Stops when	In the four cases
2. Environment and verifier	Do actions change the intended state, and is every score rebuilt from sealed evidence?	invalid actions mutate state, an oracle disagrees, hidden state leaks, replay differs	Housing v14 has 1,764 scored completions and 36 failure exclusions: completed receipt scores replay from sealed events, and failure rows are verified as exclusions; the headline measure is computed outside the receipts
3. Construct and baselines	Does a better-informed policy do better, and do shortcuts lose?	stronger and weaker policies indistinguishable, a shortcut drives the score	pay-or-test v1 fails: Procurement's fixed rules give up $45.7 to $146.1 a world, but a one-step comparison reaches the best policy to within $0.06; v2, built after the runs, passes for scripted policies: every screened rule, the one-step rule included, gives up at least $10 in every world; Refund V2.5 fails: under a rule committed before its controls ran, naive controls come too close to the reference in three of four seats
4. Attribution and controls	Does the result support the attribution it claims?	an undeclared control changes, the matrix is incomplete, the block cannot carry the claim	the three v14 identities differ in model and route and also in reasoning tier, sampling and cost ceiling, a declared choice the contract did not state; data-center claims name their counterparts
5. Confirmatory reliability and publication	Was the comparison sized, frozen and executed without tuning on outcomes?	incomplete pilot, a frozen control changed, selective missingness, a guardrail no behaviour can fail	Housing v14 has no sealed sizing record. The agreement confirmatory completed pilot, sizing, freeze and execution; its status is `partial` pending closure of analysis-digest and exclusion-verification discrepancies (F.5)

F.2 Checks applied to the four cases

Table F.2. What each kind of verifier must be shown to do, and how the four cases show it.

Kind	Must be shown	How it is shown in the four cases
Answer	the key is correct and complete for every admitted case	Refund: admission requires the policy's public compiler to reproduce the hand-written reference resolution on every case
Rule	every rule is stated to the subject, and each check fails on a counterexample	Procurement tests most of the rules its brief states against the environment; the agreement case tests its stated walk-away cost; two checks (the agreement's oracle grade and Housing's ceiling) have mutation tests; a rule left unstated or a schema that disagrees with its parser is a defect
Value	the reference agrees with the environment, and a simple rule does not already match it	each reference is played on every world (below); Procurement computed its one-step rule before play, and that rule matches its best policy
Relative	the counterparts are named and the gap is reported per counterpart	the data-center counterpart table (Table E.3)
Rubric	the rater agrees with people and resists gaming	not used

- **Regeneration.** In all four cases a test reproduces every committed pack byte for byte from its generator and seeds.
- **Reference run on every world.** Tests following initial development play each reference through the real environment and verifier with no model: the Housing reference and a full-information tenant on all 600 v14 cells, reproducing the published per-cell values and reaching the ceiling in every cell (offline, against the same functions that wrote the bundle); Procurement's best policy equals the environment's average over the 27 type profiles in all 24 worlds; the data-center oracle pair loses nothing in all 116 worlds of

both development packs, against an optimum the test computes independently; a 120-negotiation scripted dry run covered all 60 admitted agreement worlds before the confirmatory; Refund V2.5's release requires the reference to take full credit on every case.

- **Replay.** The kernel's replay recomputes each receipt's own score from its sealed events with no provider call: the 1,764 completed Housing v14 receipts (the other 36 rows are verified operational-failure exclusions); development agreement receipts replayed in-line; all 1,295 scored receipts from the agreement dry run, pilot and three confirmatory arms replayed offline. One failure receipt was checked separately for digest and seal because the pinned driver cannot classify exclusions. The procurement panels' receipts, in-line; and Refund V2.5's 816 provider-free trajectories (unpublished; 174 are distinct). The certificate does not cover two headline measures. Housing's seat-0 measure is case code over each cell's outcome, the receipts' own score totals all six tenants, and the measure was re-derived from the sealed event logs by the derived analyses of Table 5.2, re-extracted and regenerated exactly on 16 October. Procurement's original report read an unsealed action list through an unpinned reference. The 16 October reanalysis audits all 144 receipts (96 final-brief and 48 earlier-brief), reads parsed actions from their sealed logs, and pins the extraction, reference, report and case inputs. Every per-world loss and the Table E.14 intervals reproduce. This is a separately verified derivation; decision loss has not been inserted into the original receipts. The Housing receipts also name the model for the five tenant seats a script played; the true seat map is in an unsealed log.
- **Admission before types.** After development and before the confirmatory, the data-center agreement gained a pack admitted by its own rules: 60 worlds from 2,729 scanned, each admitted on its public parameters and the type prior while its hidden charge does not yet exist, then given types drawn independently of its cell (χ^2 p 0.46; the same test rejects the development packs' pattern). Tests play every shortcut through the environment for every type pair and confirm the loss admission assigned, and the oracle pair loses nothing in all 60 worlds. A separate 12-world pilot and the 60-world confirmatory have now run; the pilot supplied the sealed sizing record. This is evidence of play and variance, not a completed headroom verdict across all controls. The client seat still fails the dependence-on-counterpart-type rule.
- **Visibility.** Named hidden fields are tested absent from the observations in Housing, Procurement and Refund; the agreement case has no such test.
- **Paired plans.** A Housing test rebuilds the six v14 plans that ran, matches their ids, and asserts that they differ only in model and route, reasoning tier, sampling and cost ceiling, the last three being choices the contract did not state; Refund tests that a plan has a single subject seat; the agreement case tests its packs.
- **Golden plays.** For Housing, tests added after the runs play five kinds of tenant on v14 worlds through the real market: a good lease with its accounting checked by hand, a poor blind lemon signing, invalid and unauthorised actions (the market ends exactly as if the tenant had passed), malformed output through the kernel (typed as an error and scored as a pass, never as $0), and a world whose ceiling is $0; each has a counterexample that makes it fail.

F.3 Reading trajectories

Housing v14 (60 of its 1,800 attempts step by step) and the agreement's development runs open in a trajectory viewer that shows each step as the seat saw it beside the state it could not see, with a QC checklist per run; on 2026-10-13 that checklist flagged 670 lines across 144 bundles, of which 58 needed a reviewer. Walking a single case by hand found Refund's payment-leaf defect (Table F.4); an earlier data-center design was traced the same way on 2026-09-14.

F.4 Adversarial checks applied

- **Trivial agents** are reported beside the subjects: signing the draft unchanged recovers 14% of the value available in the development agreement pack; always paying the premium gives up $75.2 a world in Procurement; in Housing, scored after the runs, passing earns $0, signing without inspecting −$80 and an inspect-then-sign rule $145 a world, a rule two of the three models do not beat (Table 5.2).
- **Fixed rules** are scored exactly against Procurement's best policy; the one-step rule's $0.06 is the finding that fails Gates 1 and 3. A v2 pack built after the runs adds one mechanism: new listings of one seller kind share a source with a stated chance, so a cheap test of one listing tells the buyer about another. Admitted on what a buyer sees, with types drawn afterwards, its 24 worlds (from 5,798 seeds) make the one-step rule give up $30.09 a world and at least $10 in each, as do the four fixed rules and a shortcut that ignores seller kinds; with the source never shared, the one-step rule comes within $0.33 of the best policy, so the gap is the shared source. Tests check every value against the environment's average over the 27 type profiles. Its separate 12-world gate/pilot used 50 cells ($0.2193). Gemini never tested a newcomer and matched the premium rule; Luna touched one in 1 of 24 pilot plays, and neither model reached the planned probe. Required sizes were 326, 3,464 and 2,047 worlds, beyond the 96-world cap, so no confirmatory run was frozen or played.
- **Brief audits** found instructions that decided outcomes in Procurement and a schema that contradicts its verifier in Refund.
- **Lower controls for Refund V2.5** were scored under a beatability rule committed before any control ran: per seat and cohort, the reference must beat each control by at least 0.50 in end-to-end score (0.25 on policy meaning in the policy seat), and a control may match the reference on at most 10% of cases. Nine controls (empty, null, full-refund and random legal policies; an over-disclosing and a first-record customer; a first-sorted intake; a payments seat that moves nothing) played every public case, 3,468 trajectories with no model. The reference fails the rule on all 11 public cohorts in the policy, customer and payments seats: a full refund that collects nothing is fully right on 38 of 180 policy-seat cases, the evidence leaf scores any legal path as complete, the customer seat reduces to copying and the payments seat to how often the answer is no payment. Only intake passes.
- **A placebo outcome** was run after the agreement's look-back price signal: the same regression on the opening price of a scripted integrator that is never told the hidden type gave 0.38 [0.23, 0.52] where the answer must be 0, which exposed hidden types tied to the pack cell; a random relabelling of types had not caught it.

F.5 Statistical practice applied

The world is the unit of uncertainty, paired across conditions. On the agreement runs, treating plays as independent made intervals 1.47 to 1.65 times too narrow, and more worlds beat more repeats: a $50k gap needs 83 to 91 negotiations an arm at one play a world against 159 to 162 at three. Counterparts are named and reported per type: in the agreement case the counterpart accounts for 25% to 33% of the spread in what a model earns. Three reported analyses were declared before their runs: Housing v14, the agreement signalling test and the agreement confirmatory. A harness setting was tested rather than assumed: asking the CLIs for reasoning summaries changed nothing detectable in Housing (Opus +17 [−28, +62], Sol +3.5 [−46, +53]; five selected worlds, 120 episodes, the summary-off plays always run first).

Agreement confirmatory review (16 October). Gate 5 is `partial`. A 12-world variance pilot, a sealed sizing record (`aeread.design.sizing`), frozen plans, paired world-level intervals and Holm correction were applied. Eight departures are documented: CLI versions; an added wall-time stop; run-plan labels; the pack-module change for pilot generation; the permitted sizing-code copy; a post-data analysis-label fix; one missing secondary-arm negotiation; and pilot pairing by play index. The label fix changes no estimate but leaves the

analysis digest different from the sizing record; the pinned verifier still needs manual treatment of the excluded receipt. These publication and identity checks are not fully closed. The client-seat construct limit remains separate. The 45-world P2 estimate rejects zero (Holm p 0.0017), while its 98.3% interval [6.7, 46.1] lies below the declared $50k minimum; both readings must be reported.

F.6 Failure records and review

Discovered failures are recorded in the incident log in four categories (design, operational, tooling, judgment), and rows are never deleted (Table F.3). Evidence review by verification (digests, replay, prohibited-text scan) was done for the Housing v14 bundle. The agreement family's registry change has had no review by anyone other than its author, and eight earlier changes to the shared runner were accepted without one under a one-time exception.

Table F.3. Incident-log entries about each case and its predecessors, by category (as of 16 October 2026).

Case	Design	Operational	Tooling	Judgment	Total
Housing	9	21	9	12	51
Procurement	27	6	19	5	57
Data center	37	22	22	9	90
Refund	12	0	8	3	23

Counts are distinct entries. One Data-center entry had been filed under a number already in use; it is counted separately.

F.7 What quality control found

Table F.4. Twenty-five selected defects, the level at which each was visible, and what changed (dispositions as of 16 October 2026).

Case	What was wrong	Visible in one world?	How it was found	What changed
Housing	A seat-number tie-break and scripted rivals' $1 overbid decided who signed a sound listing	no: a pattern across seats	comparing signing rates by seat	one deciding tenant against declared outside demand
Housing	The scripted landlord's accept-or-counter reply was a free, near-perfect quality test beside a $25 inspection	no: needs the cost distribution	validity audit before a second model ran	pooled landlord; the revealing landlord kept as a declared arm
Housing	Admission made a favourite's lemon rate depend on its popularity (0.29 among admitted worlds where six tenants shared it, about half before admission)	no: a property of the admitted set	comparing pre- and post-admission rates	admit on the tenant's structure, read quality afterwards
Housing	The rule the tenant was told sat outside the sealed request	yes, by reading the sealed bytes	known at build, logged before any run	the rules sealed in the prompt, the tenant's history in the observation
Housing	A gap decomposition's "luck" part was not luck: uncontested blind bets were lemons 11 of 15 times, contested 1 of 15	no: a split by rival behaviour	testing the stated odds by model (p = 0.006)	blind signings scored at their expected value given what the tenant saw
Procurement (an earlier design)	A rule reading only the displayed price beat the qualified model by $28.50 a world	no: a baseline comparison	the construct check	construct gate recorded `failed`; Gate 0 made a profile mandatory

Case	What was wrong	Visible in one world?	How it was found	What changed
Procurement	The prompt pointed every subject at the cheapest listing, right in half the worlds and wrong in the other half	no: needs the pack's construction	asking whether the hard half was valid or a trick	a neutral prompt and a new pack
Procurement	The brief did not say a quote is final or that the buyer is judged on its average	no: a pattern in loss by action	listing each loss by the action that caused it	a second brief; loss fell from $12.2 to $1.0 and $19.4 to $5.4 a world
Procurement	Under sample noise the pre-award check reveals the true yield for one free action	yes, by reading the tool's code	designing a sample-again decision	open; 1 of 260 noisy cells used it, no published number moves
Data center (predecessor case)	Walking away came within $11k to $46k of the right contract, and asking for every protection within $26k to $88k	no: a strategy-class simulation	simulating shortcut policies on the pack	before any freeze: each named shortcut must cost $75k more than the best contract, and a deal must beat the better outside option by $150k
Data center (predecessor case)	At low effort the case could not separate two models: about 83% of the per-world gap was seed noise	no: needs paired seeds	variance split of the paired gap	one seed per world and more worlds; a redesign of the case
Data center	The scripted style hashed the whole world, including the client's hidden charge	no: needs two copies of a world	a test that the opening is identical for both types failed	the declared test uses a fixed style; the admitted pack draws the style from the public world only
Data center	The hidden type was fixed by pack cell and moved with public quantities	no	a placebo outcome, after the look-back (Appendix F.4)	an admitted pack whose types are drawn after admission (Appendix F.2)
Data center	The generator priced the client's rival offer, a public number in both briefs, from the drawn integrator type, so a public parameter depended on both hidden types	yes, by reading the generator	building the admitted pack	the admitted pack prices it before any type is drawn; the development packs keep theirs
Data center	The integrator's private cost barely moves the best agreement, so the client seat has almost nothing to infer about clauses	no: a value of information across worlds	building the admission rule for both seats	open; the client seat needs worlds where the integrator's risk charge is private
Housing	The primary measure conditions on the landlord's reply only below the sound floor; a reply that proves a listing sound is still scored at the stated odds	yes, by reading the measure's code	writing the formal appendix of this paper	recomputed from the sealed run roots: 57 signings move, no declared verdict changes, Gemini's leads shrink by $10 and $13 in the v14 cells; an erratum is owed
Refund	The 120-cell panel was six cases: the world seed changed only the case id, and the policy seat ran at temperature 0	no: needs the seeds compared	rebuilding the cases and reading the reference run's resolutions	the case is the unit of analysis; V2.5 varies declared conditions by seed
Refund	A correct denial written with method `null` failed: the prompt says "none", the schema allows null, the verifier compares strings	yes, by reading one proposal against the verifier	building the gap decomposition	recorded; reported as format, not outcome
Refund	The human-QC approval was minted by the runner from two status-only stubs, so it bound whatever the runner produced	yes, by reading the runner's code	the Refund audit	V2.5 names a reviewer, but the runner still binds the approval to its own output (still open)
Refund	Refund's payment leaf checks that payments executed what the proposal asked, so a proposal with an aliased mutation moves no money where the reference refunds $185.22 and still scores 1.0	yes, by tracing one case	writing the formal appendix of this paper	open; report the leaf as binding and add a check against the reference

Case	What was wrong	Visible in one world?	How it was found	What changed
Housing	The v14 receipts' own scores total all six tenant seats against baselines from a market without outside demand, so they are not the deciding tenant's	yes, by reading the receipts	building the seat-0 controls	open; the published primary is computed separately and is unaffected; a seat-0 scorer needs a module outside the hashed runner
Refund	In three of Refund V2.5's four seats a naive control comes close to the reference: a full refund that collects nothing is fully right on 38 of 180 policy cases, the evidence leaf passes any legal path, the customer seat reduces to copying and the payments seat to the base rate of no payment	no: a rate across cases	lower controls under a rule committed before they ran	open; Gate 3 recorded `failed`; scoring pairs of cases would stop the full-refund control (0 of 38 pairs), a rule not yet declared
Housing	The v14 receipts name the model for the five tenant seats a script played; the true seat map is only in an unsealed log	yes, by reading the driver's seat router	the paper's runtime review	open; record the seat map inside the seal under a new identity
Procurement	Procurement's original decision-loss report read an unsealed action copy through an unpinned reference; receipt replay alone did not certify it	yes, by reading the driver	the paper's runtime review	resolved for the reported panels on 16 October: source-pinned reanalysis from replay-audited sealed actions reproduces all 144 losses; prospective receipt-leaf integration is separate work
Procurement	The pay-or-test action schema let `message` be null while the parser requires it, so a model's first action could end its episode on format	yes, in one gate cell	the flash-model gate	the schema requires every field the parser needs, under new identities

Fourteen of 25 defects required evidence beyond one world. With Procurement's decision-loss report now re-derived from sealed actions, eight remain open: the free yield check in noisy Procurement packs; the Housing reply measure, the Housing receipts' score totals and their seat map; Refund's payment leaf, its runner-minted approval and its lower controls; and the Data-center client seat.

F.8 Assessment against the Agentic Benchmark Checklist

The four cases are assessed item by item against the checklist (Zhu et al. 2025); the outcome-validity item that applies is the one for a quality measure, since no case is scored by string matching, unit tests or a judge model.

Table F.5. The four cases against the checklist.

Item	Status	Evidence
T.1 Tool versions specified	partly	routes, prompts, schemas and implementation digests are sealed in each plan; the two CLI agents expose no seed or temperature
T.2 Required APIs accessible	partly	provider outages lost cells (36 Housing cells to an empty account, 40 data-center negotiations to CLI errors), each recorded as typed missingness
T.3 Terminates or handles API errors	partly	Housing v14 uses worker-level failure streaks: twelve workers could still produce 36 account failures. Procurement flash panels declare retries and a consecutive-failure stop; the confirmatory agreement driver freezes a five-consecutive-failure halt plus list-price and wall-time ceilings, unlike its development predecessor. These are driver controls, not adoption of the kernel halt helper (Appendix H.4); operational failures remain exclusions
T.4 Residual state cleared between runs	met	each cell is a fresh episode from a sealed plan; the CLI agents run one ephemeral subprocess per action

Item	Status	Evidence
T.5 Agent isolated from ground truth	partly	named hidden fields are tested absent from observations in Housing, Procurement and Refund, not in the agreement case; one leak was fixed and one is open in noisy-sample packs
T.6 Setup does not change over time	partly	environment inputs and declared routes are pinned; this does not attest a stable hosted deployment, provider compliance, or the unsealed CLI environment (Appendix H.2, H.3)
T.7 Ground truth verified	partly	each reference is played on every admitted world through the environment and reaches its declared result, initially by tests added after development; the admitted agreement pack also had a pre-confirmatory dry run (Appendix F.2)
T.8 Each task verified solvable	partly	reference policies are played through the environment, but full-information references can use information unavailable to the evaluated seat; this does not establish that every seat can attain the reported bound
T.9 Oracle solver	partly	exact for Procurement; an independent reference resolution and a scripted policy that resolves every case for Refund; a reachable reference for Housing, with an exact best informed tenant in development; a full-information stand-in for Data center, whose per-turn best informed policy is not built
T.10 Free of exploitable shortcuts	partly	trivial agents, exact fixed-rule scores and brief audits (Appendix F.4); twenty-five selected defects (Table F.4); a one-step rule reaches the best policy on Procurement's earlier pack, and the reported pack admits only worlds where it pays the premium; a wrong refund can still pass Refund's payment leaf
O.i.1 Quality metric resists exploitation	partly	code computes the measures and controls are tested, but the Housing measure needed correction and Refund has open payment and lower-control failures
R.1 Open source	partly	the environments, runner and evidence are public; the trajectory viewer is private
R.2 Open evaluation harness	met	the shared runner is in the public repository
R.3 Measures against contamination	met	preventive measures, not proof of zero contamination: worlds are generated, holdouts come from a disjoint seed range, and declared tests run on fresh worlds
R.4 Plans to update	partly	packs regenerate from new seeds; no refresh schedule is stated
R.5 Capability related to construct	met	each case states the decision it measures (Table 5.1)
R.6 Evaluation subject stated	met	the subject is a profile (model, harness, prompt, tools, reasoning, retries); CLI agents are named with their harness
R.7 Steps to prevent, find and correct flaws	partly	the gates and the incident log (Appendix F.1, F.6)
R.8 Qualitative discussion of unavoidable flaws	met	the end of §5.2
R.9 Quantitative analysis of flaw impact	partly	measured for the brief change, the refused cells and missingness bounds; not for every open row
R.10 Statistical significance	met	paired world-level intervals; Holm correction across declared primaries
R.11 Guidance on interpreting flawed results	met	every result is labelled confirmatory, declared or development
R.12 Non-AI baselines	partly	scripted policies in every case; no human
R.13 Trivial agents	partly	always paying (Procurement), signing the draft (Data center) and passing, signing anything and inspect-then-sign (Housing, scored after the runs) are reported; Refund V2.5 has nine lower controls, which come too close to the reference in three of four seats, and the V2.1 results reported here have none

Appendix G. Artifacts

Everything the numbers in Sections 5 and 6 rest on is in the public AERead repository or is named here when it is not.

- **Code and cases.** The shared runner and the case families, with each case's generator, reference, verifiers, packs and tests. Each case is on its own development branch of the repository.
- **Evidence.** Sealed, digest-bound bundles under `evidence/`: the Housing v14 confirmatory and its declared fill, the data-center agreement bundles, and the Refund V2.1 bundles with their gap decomposition. The Procurement pay-or-test panels are development runs kept as sealed run roots and are not yet published as bundles.
- **Quality control.** Each family's QC profile and incident log, kept with its code; the reference-run tests named in §6.
- **Sources for every number.** The source files of Sections 5 and 6 end with tables that map each table and claim to its bundle, run, code revision or log entry.
- **Paper analyses.** `tools/paper/` holds the rescoring under the corrected Housing measure, the failure taxonomy of §5.2 and the control contrasts of Table 5.2, with a runner and the inputs each needs. The Housing v14 and Procurement inputs are sealed run roots not yet published as bundles, so those numbers can be checked against the run roots but not yet from `evidence/` alone.
- **Not in the repository.** The trajectory viewer is private; one-page result summaries restate numbers from the bundles above.

Headline verification. The draft's source package includes `analysis_evidence/housing_reply_conditioned/` and `analysis_evidence/procurement_reanalysis/`. The Housing analysis re-extracts its committed trajectory table from the sealed run roots and regenerates both measures exactly. Procurement's `pay_or_test_reanalysis` checks all 144 original receipts, then reconstructs decision loss from their sealed parsed actions; its public projection regenerates Table E.14 without private run roots. Source and case digests, extraction commands, tests and a patch against the analysis base are supplied with the package. Full verification of either projection against its original execution logs still requires the retained run roots.

Agreement confirmatory and Procurement v2. The source package also includes the data-center agreement's confirmatory plan, results, analysis, sizing record and source-bound review, plus the Procurement v2 pilot report and results note. The original five agreement bundles retain their sealed bytes. The Data-center QC profile and incident-log corrections are supplied as a documentation patch; no frozen driver or analysis code was edited.

Appendix H. Runtime contract and experimental coverage

This appendix describes the runtime as of 8 October 2026 and the execution paths behind the reported results, with the later agreement confirmatory and headline verification identified separately. Reported panels used different revisions; the baseline does not retrospectively upgrade earlier execution or evidence. The full record definitions, adapter inventory and review history are retained in `technical_supplement/runtime_technical_reference.html` in the editable artifact package. Appendix G identifies the experimental artifacts.

H.1 Execution contract

Families supply economic rules, observations, actions, transitions and scorers. The kernel resolves a plan, schedules play, records calls and actions, and finalizes evidence (Figure H.1). A resolved plan binds worlds and their content hashes, seat-to-profile assignments, sampling and pairing, prompts, agent settings, budgets, retry policies and implementation pins before provider execution. Hashes detect changes to the bound bytes; they do not establish that a provider obeyed a control or that a driver followed the declared seat assignment.

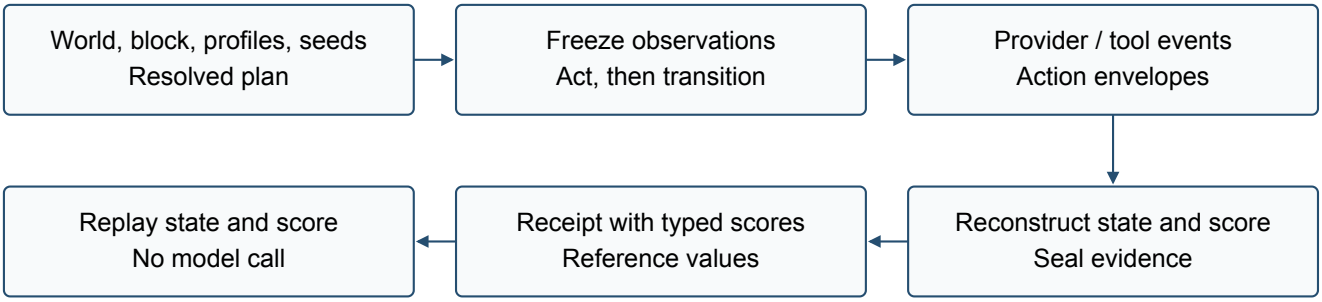


Figure H.1. The receipt path. Replay checks recorded transitions and receipt scores. It does not rerun the provider, harness or parser, or automatically cover analyses outside a receipt. The kernel supplies no sandbox or arbitrary-crash recovery.

For simultaneous play, observations are frozen before requesting any action, and the family receives the joint action set. This prevents a seat from observing a peer's same-phase action through request order; order effects inside the family transition remain the family's responsibility. Parsing and legality failures retain typed reasons. The declared invalid-action policy either stops execution or passes an invalid envelope to the family for its stated consequence. A family-hook exception is a defect in the evaluation apparatus, not a subject score.

Each attempt has an append-only event log and content-addressed artifacts. Instrumented calls record a start before dispatch and a terminal outcome; actions, transitions and termination are also recorded. Finalization reconciles these records, reconstructs scoring input, invokes the scorer and seals the evidence before writing a receipt. Retries follow the declared conditions and limits. The October path rejects empty replies and incomplete truncated answers before family parsing; older executors can differ. These guarantees cover the interfaces actually used by the driver.

H.2 Scoring, replay and access

A receipt binds the plan, world, cell, attempt, declared per-seat profile hashes, implementation pins and evidence seal to typed score records. Measurement validity and inclusion in analysis are separate statuses. Each score records its metric, reference type, scope, direction and native units. Declared applicable metrics must be returned; deferred and inapplicable metrics are listed separately. Deferred metrics cannot serve as the primary score. At most one receipt is included per planned cell, but this cardinality check cannot establish that attempt selection preceded observation of the outcome.

Replay uses the family's state-transition and scoring hooks with recorded actions and compares reconstructed states and receipt scores. It reads the stored observation, response, parse result and legality result; it does not rerun the model, harness or parser. Completed scored executions carry `state_and_score`, including those excluded as invalid measurements. Operational-failure receipts carry `none` and are verified as exclusions. Replay neither converts an exclusion into a successful episode nor certifies an analysis outside the receipt.

At the October baseline, the reviewer command is:

```
aeread verify-replay BUNDLE --run-root R \
  --setup MODULE:CALLABLE
```

It checks published bindings against retained run evidence and reports verified, differing or missing-evidence rows. Coverage extends to every published episode only when the bundle identifies that trajectory grain; otherwise it covers the receipts found under the run root. Having this command available does not establish that it was run on a given panel.

Public live bundles contain sanitized receipt projections and digests; the baseline policy omits full prompts, raw responses, model reasoning, complete live receipts and failure messages. Independent replay therefore also requires retained run roots. A digest cannot recover omitted inputs. Appendix G records access limits, and Table H.2 separates inline checks, later audits and headline-score reconstruction. Seeds and sealed requests support reconstruction of recorded play, without guaranteeing a fresh model response (He and Thinking Machines Lab 2025).

H.3 Applied controls and experimental coverage

The API and CLI paths do not apply the same controls. Table H.1 summarizes the CLI subjects in the reviewed case studies. The OpenRouter client checks reported model and route information against the declared pin, but cannot independently attest the hosted deployment or whether a requested seed or reasoning control was honored. Reasoning-effort labels are not a common scale of deliberation across models.

Table H.1. CLI controls applied in the reviewed panels. These describe the evaluated subjects, not all capabilities of the products.

Control	Claude Code / Opus 5.5	Codex CLI / Sol 6.1
Client and execution	Kernel `ClaudeCodePrintClient`; one ephemeral process per action; low effort.	External `CodexExecClient`; one ephemeral process per action; low effort.
Sampling and output	Seed, temperature and output-token limits not applied.	Seed, temperature and output-token limits not applied.
Tools and session	Tools disabled; no persistent session.	Tool definitions retained; a call using a tool is refused.
Observed identity	Version 2.1.292; binary digest `97a01e5b`, rechecked before calls.	Version 0.160.0; binary digest `112fae7a`; no served-model identity.
Accounting limits	Two internal model turns observed per kernel call; internal retries are not separate kernel attempts.	Adapter digest `3f254873`; CLI internal execution is not independently certified.
User customizations	`--safe-mode`: no CLAUDE.md, memory, skill, plugin, hook, output style or MCP server reached the model. The run's working directory, a repository checkout, and the account's email address were in context.	`--ignore-user-config` did not remove the operator's skills: eight were listed beside Codex's four built-in ones (about 500 input tokens a call), two described in benchmark and evaluation terms. The working directory's name contained the project's name.
Shared limits	Working directory and environment unsealed; costs estimated at list prices.	Working directory and environment unsealed; costs estimated at list prices.

Table H.1 describes the first agreement confirmatory. The second isolated both tools and probed their context before and during its runs, and every probe was clean (Section 5).

Table H.2 identifies the paths behind the reported results. The October executor digest prefix is `607a79ce`; the September prefix is `3cd3278c`.

Table H.2. What produced the case-study numbers. Inline replay, a later audit and headline-measure reconstruction are separate checks.

Panel	Executor, driver and client	Receipt audit and headline coverage

Panel	Executor, driver and client	Receipt audit and headline coverage
Housing v14 1,800 planned cells; separate 48-cell fill	October executor `607a79ce`; run 6 October. Family price driver; OpenRouter on pinned routes, through the seat router; minimal-chat harness.	Original panel: 1,764 completed receipts and 36 operational-failure receipts. Inline replay, then bundle replay of 600 rows per model (564 scored for Gemini), and 48 fill rows. The receipt scores six tenants. The separately versioned seat-0 reanalysis was re-extracted from sealed records and regenerated exactly on 16 October.
Housing CLI 20 worlds, 40 cells per subject	October executor `607a79ce`; `housing_cli.py`. Claude Code and the external Codex client use the same seat router.	40 completed cells per subject; inline receipt replay only. No published bundle or independent bundle-audit report in the review. Inherits Housing's seat and headline-score gaps. The driver is not digested in the plan.
Procurement CLI 24 worlds, two briefs, two subjects	October executor `607a79ce`; `procurement_cli.py`; runs on 9 October. Same CLI clients as Housing.	24 receipts per run root. On 16 October, all 96 CLI receipts across both briefs passed state and score audit; a source-pinned derivation from sealed actions reproduced every decision loss. Historical receipts still score contribution margin. Full run roots are retained.
Procurement flash 24 worlds per model	October kernel; `pay_or_test_openrouter.py`, digest `f5ec607c`. Runs 13 October; the driver bytes match its committed version. Pinned OpenRouter clients.	24 completed receipts per model; 72 and 76 provider calls. On 16 October, all 48 receipts passed state and score audit and the pinned sealed-action derivation reproduced every decision loss. Table E.14's means and intervals are unchanged.
Data-center agreement development and confirmatory	Development: September executor `3cd3278c`; `agreement_cli.py`. Confirmatory: `agreement_confirmatory_cli.py`, frozen plans, CLI versions and limits.	Development: 2,417 valid of 2,456, inline replay. Confirmatory plus pilot: 1,175 valid of 1,176; another 120 scripted dry runs. All 1,295 scored receipts replay offline; one excluded receipt has a separate digest/seal check. Earnings and IR violations are receipt leaves.
Refund V2.1 four models, 120 cells each in the reported panel	September kernel; `v2_runner`; Arena client. Contributed registration calls the shared executor and finalizer.	Published projections carry `state_and_score`; inline replay. Gap figures are regenerated by `v2_gap --check`. Produces kernel receipts despite being outside the trusted scoring-contract fixture.
Refund V2.5 816 provider-free trajectories	A Refund release on the October runtime, without kernel changes. Contributed QC path; all seats use a scripted provider; paid calls refused.	Release replays and audits all 816 receipts. No published bundle in the review. These runs qualify release conformance, not paid-model performance; they do not replace the V2.1 results.

Housing: seat attribution and headline measure.

The v14 plan names the subject profile for all six tenant seats, while the driver-side `SeatRouterClient` supplies scripts for seats 1–5. The realized routing map is retained in an unsealed side log. The original receipt scores the six-tenant sum, while the headline concerns seat 0. The original headline reads outcome facts copied into a separate unsealed result file. The later `price_reply_posterior` analysis derives its corrected measure from sealed events. On 16 October, its trajectory table was re-extracted from the original sealed run roots and the committed analysis regenerated exactly. This remains a post-run, separately versioned derivation; the original receipt's certificate is unchanged.

Procurement: the decision-loss reference.

The receipt scores contribution margin and regret against a full-information bound. Headline decision loss instead uses the optimal policy under the buyer's information. The original report replayed an `action_trace` copied into an unsealed result row through an unpinned `pay_or_test_reference`. On 16 October, a separate `pay_or_test_reanalysis` audited all 144 original receipts through state and score replay, extracted parsed actions and validity from sealed events, and recomputed every loss with a source-pinned reference. The 96 final-brief results and 48 earlier-brief results match; preserving the original bootstrap order also reproduces Table E.14's intervals. The accompanying projection permits headline recomputation without private run roots, while re-auditing the original logs still requires them. This resolves the reported panels' derivation gap. It does not add a decision-loss leaf to their historical receipts or implement the prospective case-interface standardization that has been proposed.

H.4 Scope and remaining limits

The kernel supplies neither a sandbox nor complete mediation of external activity: families and tools can run outside its ports. Resume checks and continues a valid log, but a hard kill can leave an unmatched call start that cannot be finalized. An outcome-unknown record does not prove whether an external side effect occurred. Hosted-model determinism and arbitrary-crash recovery are not claimed.

Campaign stopping and spend accounting also depend on the driver. At the reviewed baseline no family driver used the kernel's optional consecutive-failure halt helper, and there was no kernel-wide campaign cost circuit breaker. The later agreement confirmatory driver froze its own five-failure halt, list-price cap and wall-time ceiling; none fired. Housing used worker-level stop rules; twelve workers account for its 36 refused cells. The event-log spend reader distinguishes exact from incomplete totals, but had not been adopted by family drivers; CLI list-price estimates remain separate from billed charges. The inclusion rule does not prove pre-outcome attempt selection.

Integration and scientific qualification are distinct. The 21-package inventory includes wrappers, re-implementations and native packages with different fidelity, fixture and live-run coverage. The supplement retains those distinctions and the incomplete implementation-review status; package counts and passing tests do not establish case validity. Section 6 and Appendix F give the applied QC evidence and incident history. None of the four designs fully passes Gates 1–5 in this review, and later checks or repairs do not retroactively qualify earlier runs.